

Human-specific lncRNAs contributed critically to human evolution by distinctly regulating gene expression

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Abstract

What genomic sequences make protein-coding genes generate divergent expression in closely related species, specifically, differentiate humans from apes, puzzle many researchers. Many studies examined species-specific gene birth, gene loss, and changes in promoters and transcription factor binding sites, but the identification and impact of human-specific lncRNAs remain unexplored. This study identified human-specific lncRNAs from GENCODE-annotated human lncRNAs, predicted their DNA binding sites (DBSs) genome-wide, and analyzed the DBSs and their counterparts in modern humans (CEU, CHB, and YRI), archaic humans (Altai Neanderthals, Denisovans, and Vindija Neanderthals), and chimpanzees. The results reveal how human-specific lncRNAs and their DBSs have transcriptionally regulated gene expression human-specifically. The rewiring of gene expression has undergone continuous evolution, significantly changed gene expression in the brain, promoted the adaptive evolution of humans, and influenced differences in modern humans. These results reveal the importance of human-specific lncRNAs (for human evolution) and highlight the importance of other species-specific lncRNAs.

eLife assessment

In this **valuable** manuscript, the authors attempt to examine the role of long non-coding RNAs (lncRNAs) in human evolution, through a set of population genetics and functional genomics analyses that leverage existing datasets and tools. Although the methods are at times **inadequate** - for example, suitable methods and/or relevant controls are lacking at many points, and selection is inferred sometimes too quickly - the results nonetheless point towards a possible contribution of long non-coding RNAs to the evolution of human biology and they suggest clear directions for future, more rigorous study.

Introduction

Genomic changes that drive evolution include gene birth (Kaessmann, 2010), gene loss (Albalat and Canestro, 2016), and changes in regulatory sequences (Prud'homme et al., 2007). Small genomic differences between humans and other hominids, but significant phenotypic and behavioral differences, make what genomic changes at what time critically drove human evolution an enduring puzzle (Anton et al., 2014; Pollen et al., 2023). Waves of studies have identified genes that promote human brain development (Evans et al., 2005; Fiddes et al., 2018; Mekel-Bobrov et al., 2005; Pinson et al., 2022; Suzuki et al., 2018). However, conclusions from gene birth-centered studies are controversial (Currat et al., 2006; Timpson et al., 2007; Yu et al., 2007). In particular, genes important for brain development (e.g., *NOTCH2NL* and *ASPM*) do not necessarily account for many human traits that preceded or coincided with brain enlargement. Recent studies revealed the importance of regulatory sequences for human evolution, including new sequences (Liu et al., 2021), lost sequences (McLean et al., 2011), sequences showing accelerated evolution (Dong et al., 2016; Mangan et al., 2022; Prabhakar et al., 2006; Whalen and Pollard, 2022), and the turnover of transcription factor binding sites (TFBS) (Krieger et al., 2022; Otto et al., 2009; Zhang et al., 2023). An unexplored question is how simian- and human-specific lncRNAs, with the sequences of their DNA binding sites, regulate conserved genes and drive human evolution.

On this question, experimental studies have generated multiple findings. (a) The chimpanzee and human genomes are almost 99% identical (Chimpanzee Sequencing and Analysis Consortium, 2005). (b) About one-third of human lncRNAs are primate-specific (Derrien et al., 2012). (c) Species-specific lncRNAs have distinct expression profiles in organs of different species (Sarropoulos et al., 2019). (d) Many lncRNAs can bind to DNA sequences by forming RNA:DNA triplexes (Abu Almakarem et al., 2012), thereby recruiting histone and DNA modification enzymes to binding sites to regulate nearby gene expression (Lee, 2009; Roadmap Epigenomics et al., 2015). (e) Approximately 40% of differentially expressed genes in humans result from interspecies epigenetic differences (Hernando-Herraez et al., 2015). Compared with gene-centered hypotheses of human evolution, these findings suggest an alternative hypothesis – distinctly rewired gene expression by human-specific (HS) lncRNAs and their DNA binding sites has critically promoted human evolution and determined human traits.

Orthologous lncRNA genes can be identified by sequence search by structure and sequence alignment (Nawrocki, 2014; Nawrocki and Eddy, 2013). HS lncRNAs can be identified upon orthologues of human lncRNAs in other mammals. RNA:DNA triplexes follow specific base-pairing rules (Abu Almakarem et al., 2012). These rules allow the prediction of lncRNAs' DNA binding domains (DBD) and DNA binding sites (DBS) (DBDs and DBSs indicate predicted sequences of DNA binding domains and binding sites) (Lin et al., 2019; Wen et al., 2022). Gene expression in most human organs and tissues has been profiled (GTEx Consortium, 2017). Genomes of modern humans, archaic humans, and multiple apes have been sequenced (1000 Genomes Project Consortium et al., 2012; Chimpanzee Sequencing and Analysis Consortium, 2005; Meyer et al., 2012; Prufer et al., 2017; Prufer et al., 2013). These results enable us to examine the above alternative hypothesis.

HS lncRNAs and their DBSs may have occurred in three situations – (a) HS lncRNAs occurred before their DBSs, (b) HS lncRNAs and their DBSs occurred simultaneously, (c) HS lncRNAs occurred after their DBSs. It is also unknown which situation the rewiring hypothesis corresponds to. In this study, we identified HS lncRNAs upon the orthologues of the GENCODE-annotated human lncRNA genes in 16 mammalian genomes and predicted HS lncRNAs' DBDs and DBSs in the genome of modern humans (CEU, CHB, and YRI) and these DBSs' counterparts in the genomes of archaic humans (Altai Neanderthals, Denisovans, Vindija Neanderthals), and chimpanzees (**Figure**

1A). Then, we analyzed the sequences and evolution of HS lncRNAs and their DBSs, functional enrichment of target genes, and impact of HS lncRNA regulation on gene expression in GTEx organs and tissues in archaic and modern humans. Our genome and transcriptome analyses suggest that HS lncRNAs and their DBSs have substantially and distinctly rewired gene expression during human evolution. The rewiring of gene expression also greatly determines differences between human populations.

Results

1. HS lncRNAs regulate diverse genes and transcripts

After searching for the orthologues of the 13562 GENCODE-annotated human lncRNA genes (v18) in 16 mammalian genomes by sequence and structure alignment using the *Infernal* program (Lin et al., 2019; Nawrocki, 2014; Nawrocki et al., 2009), we identified 66 HS lncRNA genes (Supplementary Table 1). We predicted HS lncRNAs' DBSs in the 5000 bp promoter regions of the 179128 Ensembl-annotated transcripts (release 79) using the *LongTarget* program (Supplementary Note 1) (Lin et al., 2019). 105141 strong DBSs (binding affinity > 60, mean length > 147 bp) were identified in 96789 transcripts of 23396 genes, and 152836 weak DBSs (36 < binding affinity < 60) were identified in 127898 transcripts of 33185 genes (Figure 1B; Supplementary Table 2, 3). Genes and transcripts whose promoter regions contain the DBS of an HS lncRNA were assumed targets of the HS lncRNA. Several HS lncRNAs have DBSs in nearly all genes and transcripts (e.g., RP11-423H2.3 has strong DBSs in 73855 transcripts of 17941 genes), and only about 1.5% of target genes (0.6% of target transcripts) are human-specific (Figure 1C). Many transcripts have DBSs for multiple HS lncRNAs, many HS lncRNAs have multiple DBSs in a single transcript, and HS lncRNAs themselves show complex targeting relationships (Figure 1D; Supplementary Note 2). These findings suggest that HS lncRNAs substantially regulate gene expression.

DBS prediction was validated and supported by multiple methods and datasets (Supplementary Note 1; **Supplementary Fig 1-5**). (a) DBSs of HS lncRNAs overlap well with experimentally identified DNA methylation and histone modification signals in multiple cell lines (<https://genome.ucsc.edu>). (b) DBSs of NEAT1 and MALAT1 overlap well with experimentally identified DNA binding sites of NEAT1, MALAT1, and MEG3 (Mondal et al., 2015; West et al., 2014). (c) Many DBSs co-localize with annotated cis-regulatory elements (cCREs) in genes' promoter regions (<https://genome.ucsc.edu>). (d) We used CRISPR/Cas9 to knock out the DBD (100-200 bp) of seven lncRNAs (three HS lncRNAs and four wrongly transcribed lncRNAs) in multiple cell lines and performed RNA-seq before and after DBD knockout. Gene expression analysis revealed that the |fold change| of target genes was significantly larger than the |fold change| of non-target genes (one-sided Mann-Whitney test, $p = 3.1\text{e-}72$, $1.49\text{e-}114$, and $1.12\text{e-}206$ for the HS lncRNA RP13-516M14.1, RP11-426L16.8, and SNORA59B, and $p = 2.58\text{e-}09$, $6.49\text{e-}41$, 0.034 , and $5.23\text{e-}07$ for the four wrongly transcribed lncRNAs). The results suggest that the changed expression of target genes was caused by the knockout of DBDs. (e) Finally, the *LongTarget* program uses the Smith-Waterman local alignment to simultaneously identify DBDs and DBSs. The alignment algorithm determines that a DBS of 147 bp is extremely unlikely to be generated by chance ($p < 8.2\text{e-}19$ to $1.5\text{e-}48$).

2. Target genes characterize human evolution and traits

Since strong and weak DBSs could occur early and late in human evolution, we examined whether their target genes were enriched in specific functions important for human evolution. We sorted target genes by DBS's binding affinity, chose the top 2000 genes (DBS sequence > 252 bp, DBS binding affinity > 151) and bottom 2000 genes (DBS sequence < 60 bp, DBS binding affinity > 36), and performed over-representation analysis using the *g:Profiler* program and Gene Ontology (GO) database (Supplementary Table 4). The two target gene sets are co-enriched in GO terms important for human evolution, including "neuron projection development" (GO:0031175), "learning"

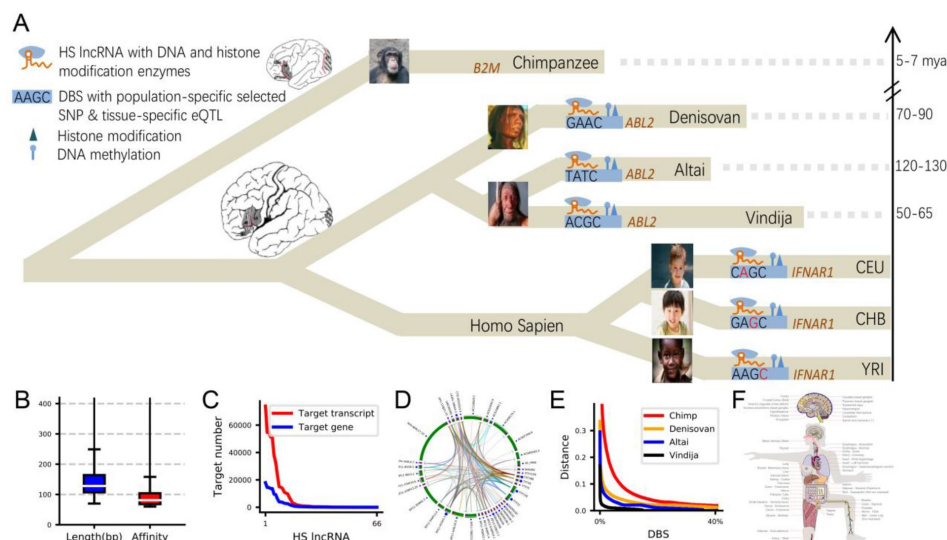


Figure 1.

Study Overview. (A) The relationships between the chimpanzees, the three archaic humans (Altai Neanderthals, Denisovans, and Vindija Neanderthals), and modern humans. The dashed lines indicate the phylogenetic distances from modern humans to archaic humans and chimpanzees. The genes and DBSs have the following implications – the DBS in *B2M* lacks a counterpart in chimpanzees; the DBS in *ABL2* has great differences in archaic humans; the DBS in *IRNAR1* is highly polymorphic in modern humans. Red characters in DBSs indicate tissue-specific eQTLs or population-specific favored mutations. (B) The distributions of DBSs' length and binding affinity. (C) The distributions of the numbers of HS lncRNAs' target genes and transcripts. (D) The targeting relationships between HS lncRNAs. (E) The distributions of sequence distances of DBSs from modern humans to chimpanzees and archaic humans. (F) The impacts of HS lncRNA regulation on gene expression in GTEx tissues.

(GO:0007612)), “regulation of anatomical structure size” (GO: GO:0090066), “head development” (GO:0060322), and “regulation of locomotion” (GO:0040012)) (Supplementary Note 3; Supplementary Table 4). 514 DBSs have binding affinity > 300 (DBS sequences > 500 bp) (Supplementary Table 5). Genes containing these strongest DBSs include *IFNAR1* and *NFATC1* (important for immune function), *KIF21B* and *NTSR1* (critical for neural development), *SLC2A11* and *SLC2A1* (involved in glucose usage), *BAIAP3* and *TAS1R3* (a brain-specific angiogenesis inhibitor and a receptor for the sweet taste response), and some primate-specific noncoding RNA genes (e.g., *CTD-3224I3.3* which is highly expressed in the cerebellum, lung, and testis) (**Table 1** [↗](#)). Interestingly, genes with weak DBSs are enriched in GO terms including “response to alcohol” (GO:0097305), “response to temperature stimulus” (GO:0009266), and “DNA methylation or demethylation” (GO:0044728), and “female pregnancy” (GO:0007565) (Supplementary Note 3; Supplementary Table 4).

3. The rewiring of gene expression has greatly evolved

The human genome is almost 99%, 99%, and 98% identical to the genomes of chimpanzees, bonobos, and gorillas (Chimpanzee Sequencing and Analysis Consortium, 2005 [↗](#); Prufer et al., 2012 [↗](#)). Given these HS lncRNA genes, we assessed to what extent their DBS sequences are human-specific (whether DBSs occurred before, at the same time, or after HS lncRNAs). 97.81% of strong DBSs were found to have counterparts in chimpanzees, suggesting that most DBS sequences occurred before HS lncRNAs.

To examine the evolution of HS lncRNAs and their DBSs during human evolution, we identified their counterparts in Altai Neanderthals, Denisovans, and Vindija Neanderthals (Meyer et al., 2012 [↗](#); Prufer et al., 2017 [↗](#); Prufer et al., 2013 [↗](#)) (**Figure 1A** [↗](#)). HS lncRNAs and strong DBSs have counterparts in these archaic humans, but the distances of HS lncRNAs are small than the distances of DBSs between archaic and modern humans (all “distance” means distance per base). We computed DBS distances in two ways – from the human ancestor (downloaded from the EBI website) to chimpanzees, archaic humans, and modern humans (the first type of distances) and between the modern humans and chimpanzees and archaic humans (the second type of distances). In the first type of distances, many DBSs’ distances from the human ancestor to modern humans were smaller than to archaic humans. For example, DBSs in the top 20% (5033) genes in the human ancestor have distances > 0.015 to modern humans, but more DBSs have distances > 0.015 to Altai Neanderthals, Denisovans, and even chimpanzees (Supplementary Note 3). This defect may be caused by that the human ancestor was built using six primates without archaic humans. The second type of distances agree well with the phylogenetic distances between chimpanzees, archaic humans, and modern humans and suggest that greatly changed sequences occurred in very limited DBSs (**Figure 1AE** [↗](#)). For example, DBSs in the top 20% (4248) genes in chimpanzees have distances > 0.034, and DBSs having distances > 0.034 in genes in Altai Neanderthals, Denisovans, and Vindija Neanderthals are 1256, 2514, and 134 (Supplementary Table 6).

To examine whether the target genes of the mostly changed DBSs in chimpanzees and archaic humans are enriched in specific functions, we performed over-representation analysis for genes with using the *g:Profiler* program and Gene Ontology (GO) database. Since 4248, 1256, and 2514 genes in chimpanzees, Altai Neanderthals, and Denisovans have DBS distances > 0.034 (to modern humans), to make the statistical significance comparable, we examined the top 1256 genes in each analysis. The two sets of genes with largest DBS distances from modern humans to chimpanzees and Altai Neanderthals are enriched in many common GO terms, but “behavior” (GO:0007610) and “cognition” (GO:0050890) have higher significance in Altai Neanderthals than in chimpanzees (**Figure 2** [↗](#)). In addition, only genes in Altai Neanderthals are also enriched in such GO terms as “learning” (GO:0007612), “learning or memory” (GO:0007611), and “glucose homeostasis” (GO:0042593). The 134 genes whose DBS distances > 0.034 from Vindija Neanderthals to modern humans, as expected, are enriched only in two GO terms with low significance, including

Target gene	Annotation	Binding affinity	Mostly changed
IFNAR1	i.e., Interferon Alpha And Beta Receptor Subunit 1.	794	C,D
NFATC1	A TF that induces gene transcription during immune responses.	736	C
NFATC1	A TF that induces gene transcription during immune responses.	491	C,A,D
ANKLE2	Diseases associated with ANKLE2 include microcephaly.	527	C,D
SEMA4D	Regulating phosphatidylinositol 3-kinase signaling, neuron projection development, and phosphate metabolic process.	495	C,A,D
KIF21B	Essential for neuronal morphology, synapse function, and learning and memory.	471	C
ALDH3B2	An aldehyde dehydrogenase for alcohol metabolism.	444	C
NTSR1	A brain and gastrointestinal peptide that mediates functions of neurotensin (e.g., hypotension, hyperglycemia, hypothermia, and antinociception).	402	C,A,D
MC5R	A receptor for melanocyte-stimulating hormone and adrenocorticotrophic hormone.	397	C

Table 1

Genes with strongest DBSs

THEG	Specifically expressed in the germ cells and involved in spermatogenesis.	395	C,D
HERC6	In pathways including class I MHC mediated antigen processing and presentation and innate immune system.	369	C,A,D
SLC2A11	Facilitating glucose transporter.	356	C
NGEF	Playing a role in axon guidance regulating ephrin-induced growth cone collapse and dendritic spine morphogenesis.	354	C
SHC2	Involved in the signal transduction pathways of neurotrophin-activated Trk receptors in cortical neurons.	345	C,D
BAIAP3	Regulating behavior and food intake by controlling calcium-stimulated exocytosis of neurotransmitters and serotonin and hormones like Insulin.	321	C
SLURP1	A marker of late differentiation of the skin.	319	C
MLPH	Involved in melanosome transport.	307	C
TAS1R3	Responding to the umami taste stimulus and recognizing diverse natural and synthetic sweeteners.	304	C,D
SLC2A1	A major glucose transporter in the mammalian blood-brain barrier.	356	C
CTD-3224I3.3	A lncRNA highly expressed in the cerebellum, lung, and testis.	312	C,A,D

C, A, D, and V indicate that the gene has a DBS with mostly changed sequence distances between modern humans and chimpanzees, Altai Neanderthals, Denisovans, and Vindija Neanderthals, respectively. In *NFATC1*, the DBS of SNORA59B has a binding affinity of 736 and the DBSs of TTTY8 and TTTY8B have a binding affinity of 491.

Table 1 (continued)

“smoothened signaling pathway” (GO:0007224) and “regulation of smoothened signaling pathway” (GO:0008589) (Supplementary Table 7). These results reveal that critical DBS changes occurred in genes involved in brain functions and some metabolism pathways in early archaic humans.

4. Gene regulation by HS lncRNAs shows differences in modern humans

We next used the 1000 Human Genome data to analyze single nucleotide polymorphisms (SNPs with minimal allele frequency (MAF) ≥ 0.1) in DBSs (1000 Genomes Project Consortium et al., 2012). Many DBSs were found to be highly polymorphic (Supplementary Table 8). The most SNP-rich DBSs (> 10 SNPs, 5 times the average number of SNPs for all DBSs) are in genes important for human evolution, including *IFNAR1*, *NFATC1*, *TAS1R3*, *INS*, *ST3GAL4*, and *FN3KRP* (Table 2). These SNPs affect DBS binding affinity, no matter they are favored, neutral, or recessive. A feature of modern human diets is high sugar intake, which causes non-enzymatic oxidation of proteins (i.e., glycation) and deactivates protein functions. Among proteins encoded by genes with the most SNP-rich DBSs, *TAS1R3* recognizes diverse natural and synthetic sweeteners, Insulin decreases blood glucose concentration, *ST3GAL4* regulates protein glycosylation, and *FN3KRP* deglycates proteins. It worth noting that DBSs in these genes have counterparts with mostly changed sequences in chimpanzees and archaic humans (Table 2), indicating human-specific evolution of the regulation of these genes.

We further used multiple statistical tests, including XP-CLR (Chen et al., 2010), iSAFE (Akbari et al., 2018), Tajima’s D (Tajima, 1989), Fay-Wu’s H (Fay and Wu, 2000), the fixation index (Fst) (Weir and Cockerham, 1984), and linkage disequilibrium (LD) (Slatkin, 2008), to detect selection signals in HS lncRNAs and strong DBSs in the human population CEU, CHB, and YRI (the representative populations of Europeans, Asians, and Africans). These tests detected selection signals in several HS lncRNA genes in CEU and CHB but not in YRI and selection signals in more DBSs in CEU and CHB than in YRI (Supplementary Note 4, 5; Supplementary Table 9, 10, 11). Examples include the DBSs in the two pigmentation-related genes *MC1R* and *MFSD12* and the two odor reception-related genes *OR6C1* and *TAS1R3*, which may reflect the adaptation of Eurasians to new environments and lifestyles.

5. SNPs in DBSs have cis-effects on gene expression

To examine the impact of DBS sequence differences on gene expression, we assessed the expression quantitative trait loci (eQTLs) in DBSs using the Genotype-Tissue Expression (GTEx) data (GTEx Consortium, 2017). 1727 SNPs with MAF ≥ 0.1 in at least one population and absolute values of cis-effect size (ES) ≥ 0.5 in at least one tissue were identified in DBSs in autosomal genes (Supplementary Table 12,13). These eQTLs include 372 “conserved” ones (in DBSs in the three archaic and three modern humans) and 1020 “novel” ones (in DBSs only in the three modern humans). Many conserved eQTLs are in the brain and have high derived allele frequencies (DAFs) in all three modern populations. A notable eQTL that has high DAFs in all three populations and positive ES ≥ 0.5 in 44 out of 48 GTEx tissues is rs2246577 in the DBS in *FN3KRP*, whose encoded protein deglycates proteins to restore their function (Table 1). On the other hand, many novel eQTLs are in a particular tissue and have population-specific DAFs (Supplementary Note 6).

We performed two analyses to examine that eQTLs in DBSs influence gene expression due to lncRNA-DBS interactions. First, we computed the expression correlation between HS lncRNAs and target transcripts with DBSs having eQTLs in specific tissues. Most (94%) HS lncRNA-target transcript pairs showed significant expression correlation in the tissues the eQTLs were identified ($|\text{Spearman's } \rho| > 0.3$ and $\text{FDR} < 0.05$). Second, we examined whether eQTLs are more enriched in DBSs than in promoter regions by computing and comparing the density of eQTLs in each DBS

GO terms	Adjusted_p	GO terms	Adjusted_p
regulation of vesicle-mediated transport	2.78E-04	behavior	1.32E-06
maintenance of location	7.46E-04	trans-synaptic signaling	1.43E-06
mitochondrial RNA metabolic process	9.22E-04	learning	2.16E-05
positive regulation of GTPase activity	1.12E-03	neuron projection development	9.91E-05
protein localization to plasma membrane	2.20E-03	cognition	1.49E-04
modification-dependent macromolecule catabolic process	2.35E-03	regulation of lipid localization	2.40E-04
regulation of GTPase activity	2.77E-03	learning or memory	4.74E-04
ubiquitin-dependent protein catabolic process	2.89E-03	response to hypoxia	5.76E-04
miRNA metabolic process	3.29E-03	associative learning	7.42E-04
receptor-mediated endocytosis	3.50E-03	regulation of neuron projection development	1.63E-03
neuron projection development	3.78E-03	response to virus	2.25E-03
behavior	5.23E-03	brain development	3.10E-03
brain development	7.47E-03	cell morphogenesis involved in neuron differentiation	3.93E-03
regulation of anatomical structure size	7.79E-03	response to nutrient levels	5.76E-03
negative regulation of neuron projection development	8.50E-03	neuron projection morphogenesis	6.68E-03
head development	8.61E-03	glucose homeostasis	6.86E-03
regulation of neuron projection development	9.23E-03	head development	7.90E-03
cell morphogenesis involved in neuron differentiation	1.07E-02	visual behavior	1.04E-02
axon development	2.96E-02	glucose metabolic process	1.41E-02
retina development in camera-type eye	3.12E-02	axon development	1.66E-02
response to starvation	3.12E-02	response to light stimulus	1.69E-02
regulation of lipid localization	3.78E-02	regeneration	1.71E-02
lipid storage	3.78E-02	sensory perception of mechanical stimulus	2.38E-02
adult locomotory behavior	4.81E-02	sensory perception of sound	2.98E-02
cognition	4.99E-02	response to alcohol	4.31E-02

Figure 2.

1256 target genes whose DBSs have largest distances between modern humans and chimpanzees and Altai Neanderthals are enriched in different *Biological Processes* GO terms. Upon significance threshold = 0.05 (Benjamini-Hochberg FDR), the two sets of genes in chimpanzees and Altai Neanderthals are enriched in 199 and 409 GO terms (50 < terms size < 1000), respectively. Pink color indicates shared GO terms. Left: Top GO terms enriched in genes in chimpanzees. Right: Top GO terms enriched in genes in Altai Neanderthals.

Target gene	Annotation	SNP number	Mostly changed
IFNAR1	i.e., Interferon Alpha And Beta Receptor Subunit 1.	31	C,D
DECR2	The related pathways include metabolism and regulation of lipid metabolism.	17	C,A,D
DOK7	Essential for neuromuscular synaptogenesis.	17	C,D
TAS1R3	Responding to the umami taste stimulus and recognizing diverse natural and synthetic sweeteners.	17	C,D
NFATC1	A TF that induces gene transcription during immune responses.	16	C
ELSPBP1	Binding to spermatozoa and playing a role in sperm capacitation.	15	C,D
ST3GAL4	Involved in protein glycosylation.	15	C,D
CAMK2B	Calcium/calmodulin-dependent protein kinase important for dendritic spine and synapse formation and maintaining synaptic plasticity.	13	C,D
HLA-DQB1-AS1	Highly expressed in EBV-transformed lymphocytes, lung, spleen.	13	C,A,D,V
ANKLE2	Diseases associated with ANKLE2 include microcephaly.	12	C,D

Table 2

Genes with the most polymorphic DBSs

KRTAP1-3	The KAP proteins form a matrix of keratin intermediate filaments that contribute to the structure of hair fibers.	12	C,A,D
INS, INS-IGF2	Insulin decreases blood glucose concentration.	11	C,A,D
SHC2	Involved in the signal transduction pathways of neurotrophin-activated Trk receptors in cortical neurons	11	C,D
FN3KRP	Deglycating proteins to restore their function, important for modern humans adaptive to high glucose intake and functions in all tissues.	10	C,D
TFB1M	The encoded protein is part of the basal mitochondrial transcription complex and is necessary for mitochondrial gene expression.	10	C,A,D

C, A, D, and V indicate that the gene has a DBS with mostly changed sequence distances between modern humans and chimpanzees, Altai Neanderthals, Denisovans, and Vindija Neanderthals, respectively.

Table 2 (continued)

and each Ensembl-annotated promoter. The results confirm that eQTLs are more enriched in DBSs (one-sided Mann-Whitney test, $p = 0.0$) (Supplementary Note 6). Thus, many SNPs in DBSs have tissue- and population-specific cis-effects on gene expression.

6. HS lncRNAs contributed to brain evolution from archaic to modern humans

We finally examined the impact of HS lncRNAs and their DBSs on gene expression in the GTEx tissues (GTEx Consortium, 2017 [DOI](#)). 40 autosomal HS lncRNAs are expressed robustly (median TPM > 0.1) in at least one tissue and form 198876 pairs with target transcripts in all tissues. 45% of these pairs show a significant expression correlation in specific tissues (Spearman's $|\rho| > 0.3$ and FDR < 0.05). In contrast, when randomly sampling 10000 pairs of lncRNAs and protein-coding transcripts genome-wide, the percent of pairs showing expression correlation with the same significance is only 2.3%. Notably, 56% of HS lncRNA-target transcript pairs with significant correlation are present in at least one brain region (**Figure 3A** [DOI](#)), indicating substantial gene expression regulation by HS lncRNAs in the brain.

To verify the above conclusion, we analyzed nine experimental datasets (Supplementary Note 7). First, we analyzed two datasets of epigenetic studies (one examined H3K27ac and H3K4me2 profiling of human, macaque, and mouse corticogenesis, and the other examined gene expression and H3K27ac modification in eight brain regions in humans and four other primates) (Reilly et al., 2015 [DOI](#); Xu et al., 2018 [DOI](#)). 84% and 73% of genes in the two datasets have DBSs for HS lncRNAs, significantly more enriched in HS lncRNA targets than the genome-wide background ($p = 1.21\text{e-}21$ and $1.2\text{e-}56$, two-sided Fisher's exact test). Second, we analyzed two datasets of PsychENCODE studies (one examined spatiotemporally differentially expressed genes and spatiotemporally differentially methylated sites in 16 brain regions, and the other examined the spatiotemporal transcriptomic divergence across human and macaque brain development) (Li et al., 2018 [DOI](#); Zhu et al., 2018 [DOI](#)). In the first dataset, 65 HS lncRNAs were expressed in all 16 brain regions, 109 transcripts had spatiotemporally differentially methylated sites in their promoters, and 56 of the 109 transcripts had DBSs for HS lncRNAs. In the second dataset, 8951 genes showed differential expression between the human and macaque brain, and 72% of differentially expressed genes in the human brain have DBSs of HS lncRNAs. Thus, both datasets show significant enrichment for HS lncRNA regulation. Third, three studies identified several genes critically regulating cortical expansion (Florio et al., 2018 [DOI](#); Johnson et al., 2018 [DOI](#); Suzuki et al., 2018 [DOI](#)). Of the 40 protein-coding genes reported by these studies, 29 have DBSs of HS lncRNAs in their promoter regions (Supplementary Table 14). Thus, these genes are enriched with DBSs of HS lncRNAs compared with the genome-wide background ($p < 0.01$, two-sided Fisher's exact test). Finally, we analyzed two datasets of brain organoids studies. By establishing and comparing cerebral organoids between humans, chimpanzees, and macaques, Pollen et al. identified 261 human-specific gene expression changes (Pollen et al., 2019 [DOI](#)). By fusing human and chimpanzee iPS cells and differentiating the hybrid iPS cells into hybrid cortical spheroids, Agolia et al. generated a panel of tetraploid human-chimpanzee hybrid iPS cells and identified thousands of genes with divergent expression between humans and chimpanzees (Agolia et al., 2021 [DOI](#)). The 261 and 1102 genes in the two datasets are enriched with DBSs of HS lncRNAs compared with the genome-wide background ($p = 1.2\text{e-}16$ and $3.4\text{e-}74$).

We next examined the evolution of the impact of HS lncRNAs and their DBSs on gene expression in the GTEx tissues. For each DBS in an HS lncRNA-target transcript pair correlated in a GTEx tissue, we computed the sequence distance of the DBS from modern humans to archaic humans. Then, we compared the distribution of DBS sequence distances in each tissue with the distribution of DBS sequence distances in all tissues (one-sided two-sample Kolmogorov-Smirnov test). DBSs in HS lncRNA-target transcript pairs correlated in brain regions have significantly changed sequence distances since the Altai Neanderthals and Denisovans (**Figure 3B** [DOI](#)), suggesting that gene expression regulation by HS lncRNAs in the brain has experienced significant evolution and that

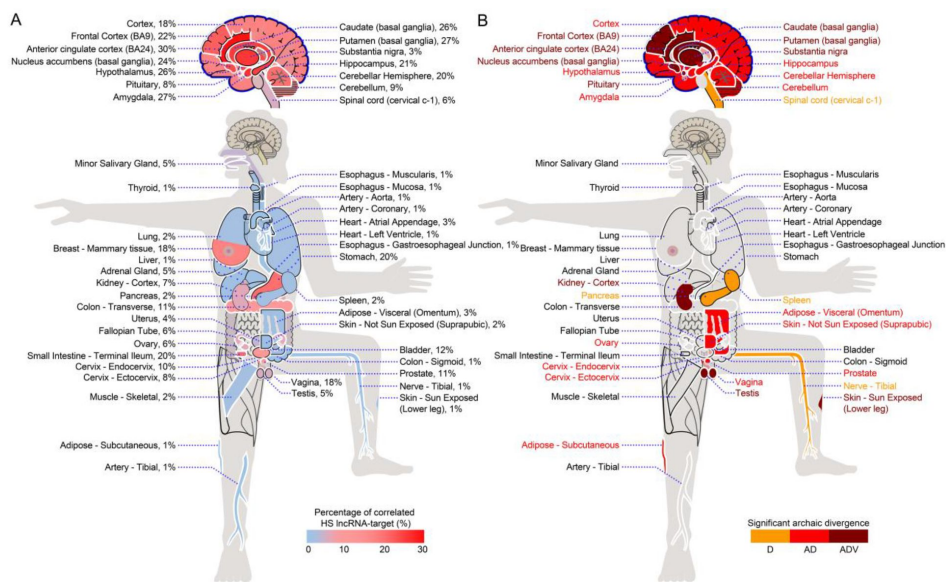


Figure 3.

The impact of HS lncRNAs on gene expression in GTEx tissues and organs. (A) Percentage distribution of correlated HS lncRNA-target transcript pairs across tissues and organs. Higher percentages of correlated pairs are in the brain than in other tissues and organs. (B) The distribution of significantly changed DBSs in HS lncRNA-target transcript pairs across tissues and organs between archaic and modern humans. Orange, red, and dark red indicate significant DBS changes from Denisovans (D), Altai Neanderthals and Denisovans (AD), and all of the three archaic humans (ADV). DBSs in HS lncRNA-target transcript pairs correlated in the seven brain regions have changed significantly and consistently since the Altai Neanderthals, Denisovans, and Vindija Neanderthals (one-sided two-sample Kolmogorov-Smirnov test, significant changes determined by $FDR < 0.001$).

HS lncRNAs contributed greatly to rapid brain evolution. Results from other GTEx tissues, including Kidney_Cortex, Ovary, and Skin, also suggest that the rewired gene expression has influenced human evolution.

Discussion

Modern humans differ from archaic humans and great apes little in genome sequences but greatly in phenotypes and behaviors, making what genomic differences critically determine the phenotypic and behavioral differences a compelling question. Several HS protein-coding genes promote the rapid expansion of neocortex (Fiddes et al., 2018 [↗](#); Florio et al., 2018 [↗](#); Pinson et al., 2022 [↗](#)). Variations in TFBS cause gene expression changes (Franchini and Pollard, 2017 [↗](#)). The rewiring of gene expression by TFs and TFBSs evolves (Johnson, 2017 [↗](#)), so should the rewiring of gene expression by lncRNAs and their DBSs. However, the rewiring and evolution of the second case have been barely examined and poorly understood. This study identified HS lncRNAs and their DBSs, analyzed the rewiring and evolution of gene expression, and presents a new answer to the question of what genomic changes at what time critically drove human evolution. We performed seven CRISPR/Cas9 experiments to validate DBS prediction and analyzed multiple experimental datasets to validate our genome and transcriptome analyses. These results support the hypothesis that HS lncRNAs have distinctly rewired gene expression for adaptive evolution of humans.

Several findings are noteworthy. First, Neanderthal-inherited sequences have measurable impacts on gene expression, but the impacts are least detectable in the brain (McCoy et al., 2017 [↗](#)). Here, we found that HS lncRNAs have more significantly influenced gene expression in the brain than in other tissues. Second, the regulation of genes by HS lncRNAs, making humans adapt to high sugar intake, has undergone great evolution. Third, genes regulated by HS lncRNAs are enriched in different functions at different stages. Fourth, old SNPs in DBSs are in genes regulating neural development, and new SNPs in DBSs are in genes regulating such processes as glucose metabolism.

The above findings help reveal the functions of HS lncRNAs and the adaptive evolution of humans and highlight the important functions of other species-specific lncRNAs. They also raise three questions. First, the protein-coding regions in the mouse and human genomes are about 85% identical, but many mouse lncRNAs are rodent-specific (Yue et al., 2014 [↗](#)). Thus, a question is how mouse-specific lncRNAs specifically rewire gene expression in mice and how the rewiring influences the transcriptional differences between humans and mice (Breschi et al., 2017 [↗](#); Hodge et al., 2019 [↗](#); Zhu et al., 2018 [↗](#)). Second, whether mice and other mammals have a lncRNA like RP11-423H2.3 that substantially rewires gene expression species-specifically. Third, the extent clade- and species-specific lncRNAs would make speciation and evolution preordained. These questions are of great interest and have practical importance.

Material and methods

1. Data sources

The sequences of human (GRCh38/hg38, GRCh37/hg19) and chimpanzee genomes (panTro5) were obtained from the UCSC Genome Browser (<http://genome.ucsc.edu> [↗](#)). Three high-quality archaic human genomes were obtained from the Max Planck Institute for Evolutionary Anthropology (<https://www.eva.mpg.de/genetics/index.html> [↗](#)), which include an Altai Neanderthal which lived approximately 122 thousand years ago (kya) (Prufer et al., 2013 [↗](#)), a Denisovan (an Asian relative of Neanderthals) which lived approximately 72 kya (Meyer et al., 2012 [↗](#)), and a Vindija

Neanderthal which lived approximately 52 kya (Prufer et al., 2017 [DOI](#)). The ancestral sequences for the human genome (GRCh37) (which were generated upon the EPO alignments of six primate species human, chimpanzee, gorilla, orangutan, macaque, and marmoset) were downloaded from the EBI website

(ftp://ftp.1000genomes.ebi.ac.uk/vol1/ftp/phase1/analysis_results/supporting/ancestral_alignments/). The SNP data of modern humans were obtained from the 1000 Genomes Project (phase I) (1000 Genomes Project Consortium et al., 2012 [DOI](#)). The three modern human populations are CEU (Utah residents with Northern and Western European ancestry), CHB (Han Chinese in Beijing, China), and YRI (Yoruba in Ibadan, Nigeria), which contain 100, 99, and 97 individuals, respectively. The genetic map files of modern humans were downloaded from the 1000 Genomes Project website (ftp://ftp-trace.ncbi.nih.gov/1000genomes/ftp/technical/working/20110106_recombination_hotspots/). The 1000 Genomes phased genotype data in VCF format were downloaded from the International Genome Sample Resource website (<http://www.internationalgenome.org/>). Annotated human genes and transcripts were obtained from the Ensembl website (<http://www.Ensembl.org>). FASTA-format nucleotide sequences were converted from VCFs using the *vcf-consensus* program in the *VCFtools* package when necessary (Danecek et al., 2011 [DOI](#)). Data in the GeneCards Human Gene Database (www.genecards.org) (<http://www.genecards.org/>) were used to annotate genes and transcripts.

RNA-sequencing (RNA-seq) data (transcripts per million [TPM]) and eQTL data of human tissues were obtained from the Genotype-Tissue Expression Project (GTEx, v8) website (<https://gtexportal.org/>) (GTEx Consortium, 2017 [DOI](#)). Histone modification and DNA methylation signals in multiple cell lines and ENCODE Candidate Cis-Regulatory Elements (cCRE) were obtained from the UCSC Genome Browser (<http://genome.UCSC.edu>). The genome-wide DNA binding sites of lncRNA NEAT1, MALAT1, and MEG3 were obtained from experimental studies (Mondal et al., 2015 [DOI](#); West et al., 2014 [DOI](#)).

2. Identifying HS lncRNA genes

We used the *Infernal* program (Nawrocki et al., 2009 [DOI](#)), which searches orthologous RNA sequences upon sequence and structure alignment, to identify orthologous exons in 16 mammalian genomes for each exon in each of the 13562 GENCODE (v18)-annotated human lncRNA genes (Lin et al., 2019 [DOI](#)). The 16 mammals were chimpanzee (CSAC 2.1.4/panTro4), macaque (BGI CR_1.0/rheMac3), marmoset (WUGSC 3.2/calJac3), tarsier (Broad/tarSyr1), mouse lemur (Broad/micMur1), tree shrew (Broad/tupBel1), mouse (GRCm38/mm10), rat (Baylor3.4/rn4, RGSC6.0/rn6), guinea pig (Broad/cavPor3), rabbit (Broad/oryCun2), dog (Broad CanFam3.1/canFam3), cow (Baylor Btau_4.6.1/bosTau7), elephant (Broad/loxAfr3), hedgehog (EriEur2.0/eriEur2), opossum (Broad/monDom5), and platypus (WUGSC 5.0.1/ornAna1) (<http://genome.UCSC.edu>). If the number of orthologous exons of a human lncRNA gene in a genome exceeded half the exon number of the human lncRNA gene, these orthologous exons were assumed to form an orthologous lncRNA gene. If a human lncRNA gene had no orthologous gene in all of the 16 mammals, it was assumed to be human-specific.

3. Identifying DBSs of HS lncRNAs

lncRNAs bind to DNA sequences by forming RNA:DNA triplexes. Each triplex comprises triplex-forming oligonucleotides (TFO) in the lncRNA and a triplex-targeting site (TTS) in the DNA sequence. We used the *LongTarget* program to predict HS lncRNAs' DNA binding domains (DBD) and binding sites (DBS) with the default parameters (*Ruleset* = all, *TT penalty* = -1000, *CC penalty* = 0, *Offset* = 15, *Identity* ≥ 60, *Nt* ≥ 50) (Lin et al., 2019 [DOI](#); Wen et al., 2022 [DOI](#)). *LongTarget* program predicts DBDs and DBSs simultaneously, with a DBD comprising a set of densely overlapping TFOs and a DBS comprising a set of densely overlapping TTSs. For each HS lncRNA, we predicted its DBSs in the 5000 bp promoter regions (-3500 bp upstream and +1500 bp downstream the transcription start site) of the 179128 Ensembl-annotated transcripts (release 79). For each DBS, a binding affinity was calculated by multiplexing the DBS length and the averaged *Identity* score of

all TTSs (the *Identity* score is the percentage of paired nucleotides). Strong and weak DBSs were classified upon binding affinity ≥ 60 and < 60 . A transcript whose promoter region contains a strong DBS of an HS lncRNA was assumed to be a target transcript of the HS lncRNA, and the gene containing this transcript was assumed to be a target gene of the HS lncRNA. As the 1000 Genomes Project (phase I) data and the archaic human genomes are based on GRCh37/hg19, the coordinates of DBSs were converted from GRCh38/hg38 to GRCh37/hg19 using the *liftover* program from the UCSC Genome Browser (Kuhn et al., 2013 [↗](#)).

4. Experimentally validating DBS prediction

A 157 bp sequence (chr17:80252565-80252721, hg19) containing the DBD of RP13-516M14.1, a 202 bp sequence (chr1:113392603-113392804, hg19) containing the DBD of RP11-426L16.8, and a 198 bp sequence (chr17:19460524-19460721, hg19) containing the DBD of SNORA59B, were knocked out in the HeLa cell line, RKO cell line, and SK-MES-1 cell line, respectively. Two sequences (chr1:156643524-156643684, chr10:52445649-52445740, hg38) containing the DBD of two wrongly transcribed noncoding sequences were knocked out in the HCT-116 and A549 cell lines, respectively. The seven cases of knocked out were performed by UBIGENE, Guangzhou, China, using CRISPR-UTM, a revised version of CRISPR/Cas9 technology (<http://www.ubigene.com> [↗](#)). The RNA-sequencing (RNA-seq) was performed by Novogene, Beijing, China (<https://cn.novogene.com> [↗](#)) and HaploX, Shenzhen, China (<https://www.haplox.cn/> [↗](#)), respectively.

5. Mapping DBSs of HS lncRNAs in the chimpanzee and archaic human genomes

We used the *liftover* program from the UCSC Genome Browser (<http://genome.UCSC.edu> [↗](#)) to map loci of DBSs in the human genome hg38 to the chimpanzee genome Pan_tro 3.0 (panTro5). The mapping results were examined by checking the human-chimpanzee pairwise alignment (hg38 vs panTro5) in the UCSC Genome Browser (<http://genome.UCSC.edu> [↗](#)). 97.81% (100403/102651) of DBSs were mapped, and the 2248 unmapped DBSs in 429 genes were considered human-specific. We used *vcf-consensus* in the *VCFtools* package to extract the DBSs of HS lncRNAs from the VCF files of Altai Neanderthals, Denisovans, and Vindija Neanderthals. The variant with the highest quality score was selected whenever multiple variant calls were observed at a given locus. The obtained DBS sequences in the chimpanzees and three archaic humans are called counterparts of DBSs in these genomes.

6. Estimating sequence distances of DBSs between different genomes

We first aligned DBS sequences in the genomes of humans, chimpanzees, Altai Neanderthals, Denisovans, and Vindija Neanderthals using the *MAFFT7* program for measuring sequence distances from modern humans to chimpanzees and archaic humans (Katoh and Standley, 2013 [↗](#)). We then computed sequence distances using the *dnadist* program with the Kimura 2-parameter model in the *PHYLIP* (3.6) package (<http://evolution.genetics.washington.edu/phylip.html> [↗](#)) and using the *Tamura-Nei model* in the *MEGA7* package (Kumar et al., 2016 [↗](#)). The two methods generated equivalent results. The largest distance between DBSs in humans and their counterparts in chimpanzees is 5.383. Since 2248 DBSs in 429 human genes do not have counterparts in chimpanzees, we assumed that the sequence distances of these DBSs between humans and chimpanzees were 10.0. Since the top 20% of DBSs in chimpanzees (including these 2248 DBSs) have distances ≥ 0.034 , we used 0.034 as the threshold to identify the “mostly changed” DBSs in the chimpanzees and three archaic humans.

We determined human ancestral sequences of DBSs upon the human ancestor sequences from the EBI website, which were generated upon the EPO alignments of six primate species. We used the above-mentioned methods to compute DBSs’ sequence distances from the human ancestor to chimpanzees, archaic humans, and modern humans. We found that when the human-chimpanzee

ancestral sequence has the ancestral sequence (which means the inferring of ancestral allele is of high confidence), DBS distances from the human ancestor to modern humans are larger than to archaic humans, but this situation only accounts for about 63.8%. For many DBSs, the distances from the human ancestor to modern humans are smaller than to archaic humans (especially Neanderthals and Denisovans) and even to chimpanzees. This defect may be caused by the absence of archaic humans in building the human ancestral sequence.

7. Detecting positive selection signals in HS lncRNA genes and DBSs

We used multiple tests to detect positive selection signals in HS lncRNA genes. First, we used the XP-CLR test (parameters = -w1 0.001 300 100 -p0 0.95, window size = 0.1 cM, grid size = 100 bp) to perform six pairwise genome-wide scans (i.e., CEU-CHB, CEU-YRI, CHB-CEU, CHB-YRI, YRI-CEU, and YRI-CHB) (Chen et al., 2010 [\[10\]](#)). The upper 1% of scores across the entire genome in each pairwise scan were 34.6 in the CEU-YRI scan, 16.8 in the CEU-CHB scan, 45.0 in the CHB-YRI scan, 26.9 in the CHB-CEU scan, 14.1 in the YRI-CEU scan, and 14.1 in the YRI-CHB scan. These scores were used as the thresholds of positive selection signals in these populations. Second, we used the *iSAFE* program to scan each genomic region that contains a HS lncRNA gene and its 500-kb upstream and downstream sequences (Akbari et al., 2018 [\[11\]](#)). Strongly selected loci were detected only in CEU and CHB. Third, we used the *VCFTools* program to calculate Tajima's D values for each HS lncRNA gene in CEU, CHB, and YRI (Danecek et al., 2011 [\[12\]](#)). The calculation was performed using a 1500-bp non-overlapping sliding window because the lengths of these genes exceed 1500 bp. To generate the background reference for judging the significant increase or decrease of Tajima's D for HS lncRNA genes in a population, we calculated Tajima's D values across the whole genome using a sliding window of 1500 bp. As values of Tajima's D were compared with the background reference, a significant $D < 0$ and $D > 0$ indicate positive (or directional) selection and balancing selection, respectively, instead of population demography dynamics (Tajima, 1989 [\[13\]](#)). Fourth, we used the integrated *Fst* to detect positive selection signals in HS lncRNA genes. The *Fst* of each HS lncRNA gene was computed for three comparisons, i.e., CEU-YRI, CHB-YRI, and CHB-CEU. Extreme *Fst* values of SNPs were detected in HS lncRNA genes in the comparisons of CEU-YRI and CHB-YRI. Since allele frequencies for different loci vary in a genome and genetic drift may have different effects at different loci, we used a sliding window of 1500 for comparing *Fst* values of HS lncRNA genes with the genome-wide background. Extreme *Fst* values indicate positive selection. Finally, we applied linkage disequilibrium (LD) analysis to each HS lncRNA gene. We computed the pairwise LD (r^2) in CEU, CHB, and YRI for common SNPs (with minimal allele frequency (MAF) ≥ 0.05 in at least one population) in HS lncRNA genes and DBSs using the *PLINK* program (Purcell et al., 2007 [\[14\]](#)). Significantly increased LD was detected in SNPs in HS lncRNA genes in CEU and CHB. The LD patterns were portrayed using the *Haploview* program (Barrett et al., 2005 [\[15\]](#)).

Next, we used the above tests to detect positive selection signals in DBSs. First, the 100 bp grid size of the XP-CLR test also allowed the detection of selection signals in DBSs. Second, we performed the Tajima's D and Fay-Wu's H tests (Fay and Wu, 2000 [\[16\]](#); Tajima, 1989 [\[13\]](#)). We calculated Tajima's D values for each DBS in CEU, CHB, and YRI using the *VCFTools* program (Danecek et al., 2011 [\[12\]](#)), with a sliding window of 147 bp (the mean length of strong DBSs). To generate the background reference for judging the significant increase or decrease of Tajima's D in a population, we calculated Tajima's D values across the whole genome using the same sliding window. As values of Tajima's D were compared with the background reference, a significant $D < 0$ and $D > 0$ indicate positive (or directional) selection and balancing selection, respectively. Fay-Wu's H values were calculated similarly using the *VariScan* program (Vilella et al., 2005 [\[17\]](#)). Calculating Fay-Wu's H demands the ancestral sequences as the outgroup. We extracted the ancestral sequences of DBSs from the human ancestral sequence which was generated upon the EPO alignments of six primate species (human, chimpanzee, gorilla, orangutan, macaque, and marmoset) (ftp://ftp.1000genomes.ebi.ac.uk/vol1/ftp/phase1/analysis_results/supporting/ancestral_alignments/). Third, we computed the *Fst* to measure the frequency differences of alleles in DBSs between

populations. For the CEU-YRI, CHB-YRI, and CHB-CEU pairwise comparisons, we used the revised *VCFTools* program to compute the weighted F_{st} for all SNPs in each DBS (Weir and Cockerham, 1984). Fourth, we integrated the weighted F_{st} values in the three populations into an “integrated F_{st} ” which indicated whether the DBS locus was under selection in a certain population (Nielsen and Slatkin, 2013). We used sliding windows of 147 bp for comparing F_{st} values of DBSs with the genome-wide background. We empirically denoted the values of integrated F_{st} in the upper 10% of scores across the entire genome as statistically significant. To determine positive selection more reliably, we used Tajima’s D and the integrated F_{st} to jointly determine if a DBS was under positive selection in a population. The thresholds which determined the upper 10% of Tajima’s D values across the entire genome in CEU, CHB, and YRI were -0.97, -0.96, and -0.97, respectively, and the threshold which determined the upper 10% of integrated F_{st} values across the entire genome in the three populations was 0.22. For example, a DBS was assumed to be under positive selection in CEU if (a) the DBS had a Tajima’s $D < -0.97$ in CEU and Tajima’s $D > 0.0$ in the two other populations and (b) the DBS had an integrated $F_{st} > 0.22$. Analogously, a DBS was assumed to be under positive selection in both CEU and CHB if the DBS had a Tajima’s $D < -0.97$ in CEU, < -0.96 in CHB, and > 0.0 in YRI, and had an integrated $F_{st} > 0.22$.

8. Functional enrichment analysis of genes

We used the *g:Profiler* program (with the parameter settings: Organism=Homo sapiens, Ordered query = No, Significance threshold = Benjamini-Hochberg FDR, User threshold = 0.05, 50 < terms size < 1000) and the Gene Ontology (GO) database to perform over-representation analysis (Raudvere et al., 2019). Over-representation analysis is used to determine which *a priori* defined gene sets are more present (over-represented) in a subset of “interesting” genes than what would be expected by chance. Here, we used the analysis to examine different target gene sets are enriched in what GO terms (*Biological Process*). The target gene sets included genes with strongest DBSs in humans, genes with weakest DBSs in humans, and genes with the mostly changed DBSs in chimpanzees and three archaic humans. To choose genes with strongest DBSs, we ranked genes in descending order based on DBS binding affinity and chose the top 2000 genes for over-representation analysis. To choose genes with weakest DBSs, we first ranked genes in ascending order based on DBS binding affinity, then ranked genes in descending order based on the number of regulatory HS lncRNAs, and finally chose the top 2000 genes with binding affinity ≤ 40 and regulatory HS lncRNAs ≥ 5 . We used the distance threshold of 0.034 to choose genes with mostly changed DBSs. Genes with mostly changed DBSs in chimpanzees include the 429 genes in which 2248 DBSs lack counterparts. 4248, 1256, 2514, and 134 genes in chimpanzees, Altai Neanderthals, Denisovans, and Vindija Neanderthals have DBSs or their counterparts with distance ≥ 0.034 . To make the statistical significance of enrichment analysis comparable across chimpanzees, Altai Neanderthals, and Denisovans, we used the same number of genes (the top 1256 genes whose DBSs have the largest DBS distances) as the input to the *g:Profiler* program.

9. Analysing SNP frequencies in human populations

The frequency of common SNPs ($MAF \geq 0.05$) in DBSs in the three modern human populations was computed using the *VCFTools* package (Danecek et al., 2011). The ancestral/derived states of SNPs were inferred upon the human ancestor sequences and were used to determine derived allele frequency.

10. Analysing the cis-effect of SNPs in DBSs on target gene expression

SNPs with $MAF > 0.1$ in DBSs in any of the three modern human populations and absolute values of cis-effect size > 0.5 ($FDR < 0.05$) in any of the GTEx tissues were examined for an influence on the expression of the target genes (GTEx Consortium, 2017). SNPs that are eQTLs in the GTEx tissues and have biased derived allele frequencies (DAF) in the three modern human populations were examined to estimate whether the eQTL is population-specific.

11. Examining the tissue-specific impact of HS lncRNA-regulated gene expression

First, we examined the expression of HS lncRNA genes across the GTEx tissues. HS lncRNA genes with a median TPM value > 0.1 in a tissue were considered robustly expressed in that tissue. Upon this criterion, 40 HS lncRNA genes were expressed in at least one tissue and were used to examine the impact of HS lncRNA regulation on gene and transcript expression (other HS lncRNAs may function in the cytoplasm). Since a HS lncRNA gene may have multiple transcripts, we adopted the transcript which contains the predicted DNA binding domain (DBD) and has the largest TPM value as the representative transcript of the HS lncRNA. We calculated the pairwise Spearman's correlation coefficient between the expression of a HS lncRNA (the representative transcript) and the expression of each of its target transcripts using the *scipy.stats.spearmanr* program in the *scipy* package. The expression of a HS lncRNA gene and a target transcript was considered to be significantly correlated if the $|\text{Spearman's } \rho| > 0.3$, with Benjamini-Hochberg FDR < 0.05 . We examined the percentage distribution of correlated HS lncRNA-target transcript pairs across GTEx tissues and organs (**Figure 3A** [↗](#)).

We examined all GTEx tissues to determine in which tissues HS lncRNA-regulated gene expression may have undergone significant changes from archaic to modern humans. If a pair of HS lncRNA and target transcript are robustly expressed in a tissue and their expression shows a significant correlation ($|\text{Spearman's } \rho| > 0.3$, with Benjamini-Hochberg FDR < 0.05) in the tissue, we computed the sequence distance of the HS lncRNA's DBS in the transcript from the three archaic humans to modern humans. We compared the sequence distances of all DBSs in each tissue with the sequence distances of all DBSs in all GTEx tissues (i.e. the distribution of all DBS distances was taken as the background). A one-sided two-sample Kolmogorov-Smirnov test was used to examine whether the sequence distances of all DBSs in a specific tissue deviate from the background distribution (which reflects the “neutral evolution” of gene expression). For a tissue, if the test result was Benjamini-Hochberg FDR < 0.001 , the tissue was considered to have significantly changed gene expression regulated by the HS lncRNA. We used different colors to mark tissues with significantly changed gene expression regulation since Altai Neanderthals, Denisovans, and Vindija Neanderthals, and used “D”, “AD”, and “ADV” to indicate changes since Denisovans, since Altai Neanderthals and Denisovans, and since Altai Neanderthals, Denisovans, and Vindija Neanderthals, respectively (**Figure 3B** [↗](#)).

Declaration of competing interest

The authors declare no competing interest.

Author contributions

H.Z. designed the study and drafted the manuscript. J.L. performed most data analyses. Y.W. and J.T. performed XP-CLR and iSAFE analyses. All authors have read the manuscript and consent to its publication.

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Data and code availability

GENCODE human lncRNAs are available at the website (<https://www.genencodegenes.org/human/>). Human lncRNAs' orthologues in 16 mammals and the *LongTarget* program are available at the *LongTarget/LongMan* website (<http://www.gaemons.net/>). Other programs and tests are open-source and freely available. The RNA-seq data of the three HS lncRNAs before and after the CRISPR/Cas9 experiments have been deposited in the NCBI GEO database (<https://www.ncbi.nlm.nih.gov/geo/>) under the accession number GSE213231. The RNA-seq data of the four wrongly transcribed noncoding sequences before and after the CRISPR/Cas9 experiments have been deposited in the NCBI GEO database (<https://www.ncbi.nlm.nih.gov/geo/>) (the accession number is to be obtained). All other data are in Supplementary Tables.

Supplementary data

Two supplementary files are available with the manuscript online. One PDF file contains 7 Supplementary Notes in which there are 23 Supplementary Figures, and one Excel file contains 14 Supplementary Tables.

Supplementary Note 1 – Supporting evidence for DBS prediction

We used multiple methods and datasets to validate DBS prediction. First, since lncRNAs can recruit epigenetic regulatory enzymes to lncRNAs' DNA binding sites, we uploaded predicted DBSs of HS lncRNAs to the UCSC Genome Browser as a custom track, compared these DBSs with tracks such as "ENCODE DNA Methylation tracks" (Meissner *et al.*, 2008) and "ENCODE Histone Modification tracks" (Ernst *et al.*, 2011). We found that predicted DBSs overlap well with experimentally identified DNA methylation and histone modification signals in multiple cell lines (**Supplementary Figure 1**).

Second, since lncRNAs' DBSs can be detected genome-wide by experiments such as ChIRP-seq, we predicted the DBSs of the lncRNA MALAT1, NEAT1, and MEG3 and compared predicted DBSs with the experimentally identified DNA binding sites (Mondal *et al.*, 2015; West *et al.*, 2014). We found that predicted DBSs agree well with experimentally identified DNA binding sites (**Supplementary Figure 2**).

Third, we found that many DBSs also co-localize with ENCODE Candidate Cis-Regulatory Elements (cCREs) in promoter regions of genes (**Supplementary Figure 3**). cCREs are the subset of representative DNase hypersensitive sites across ENCODE and Roadmap Epigenomics samples supported by either histone modifications (H3K4me3 and H3K27ac) or CTCF-binding data.

Fourth, we used the CRISPR/Cas9 technique to knock out the sequences that contain the DBD of seven lncRNAs (three HS lncRNAs and four wrongly transcribed long noncoding RNAs) in multiple cell lines and performed RNA-seq before and after the knockout. The deleted sequences were just 100-200 bp, and gene expression analysis revealed that the |fold change| of target genes was significantly larger than the |fold change| of non-target genes (one-sided Mann-Whitney test, $p = 3.1\text{e-}72$, $1.49\text{e-}114$, and $1.12\text{e-}206$ for HS lncRNA RP13-516M14.1, RP11-426L16.8, and SNORA59B, and $p = 2.58\text{e-}09$, $6.49\text{e-}41$, 0.034 , and $5.23\text{e-}07$ for the four wrongly transcribed lncRNAs in cancer cell lines) (**Supplementary Figure 4**). These results suggest that the deletion of DBD causes the changed expression of target genes (**Supplementary Figure 5**).

Fifth, the *LongTarget* program uses a variant of the Smith-Waterman algorithm to identify all local alignments in each lncRNA/DNA pair. Each alignment is a RNA:DNA triplex that consists of a TFO (triplex-forming oligonucleotides) and a TTS (triplex-targeting sites). A DBS is defined based on a set of densely overlapping TTSSs, and a DBD is defined based on both TFOs and the DBS (He et al., 2015). *LongTarget* can more robustly and accurately identify DBS than some popular methods (Wen et al., 2022). To examine the likelihood a local alignment of 147 bp (the average length of strong DBSs) can be generated by chance in a 5000 bp DNA sequence, we randomly simulated two sequences seqA and seqB (which were 5000 bp and 147 bp) and aligned seqB to seqA using the *EMBOSS Water* (https://www.ebi.ac.uk/Tools/psa/emboss_water/) local alignment program with alignment identity controlled to 60% (because the default *Identity* parameter of the *LongTarget* program was 60%). The equation $p = 1 - \exp(-Kmne^{-\lambda S})$ was used to calculate a p value to estimate the likelihood, in which m and n were the length of seqA and seqB, K was a small adjusting constant (which was 0.1), λ was the normalization coefficient of the score (which was 0.3), and S was the alignment score. We repeated the simulation and calculation process 10000 times and determined that the maximal and minimal p values were between $8.2e-19$ and $1.5e-48$. Thus, a DBS of 147 bp is extremely unlikely to be generated by chance. This result also supports that the changed expression of target genes is caused by the knockout of DBDs in the CRISPR/Cas9 experiments.

Supplementary Note 2 – Features of HS lncRNA-mediated gene expression regulation

HS lncRNA-mediated gene expression regulation shows multiple features. First, HS lncRNAs themselves form complex targeting relationships (Supplementary Figure 6), suggesting networks and cascades of regulation. Second, some DBSs are human-specific sequences (Supplementary Figure 7), suggesting that human-specific sequences were explored by HS lncRNAs for regulating gene expression. Third, many genes and transcripts have DBSs for multiple HS lncRNAs (Supplementary Figure 8), suggesting co-regulation of functionally related genes. Fourth, certain genes and transcripts have multiple DBSs for one HS lncRNA, suggesting tissue-specific regulation. Fifth, selection signals were detected in some DBSs in specific populations (Supplementary Figure 9). Sixth, generally, HS lncRNAs on the Y chromosome have longer DBSs than HS lncRNAs on the autosomes (Supplementary Figure 10). Finally, SNPs in the DBSs of a HS lncRNA in multiple genes on a chromosome show LD, suggesting an association between these DBSs (Supplementary Figure 11).

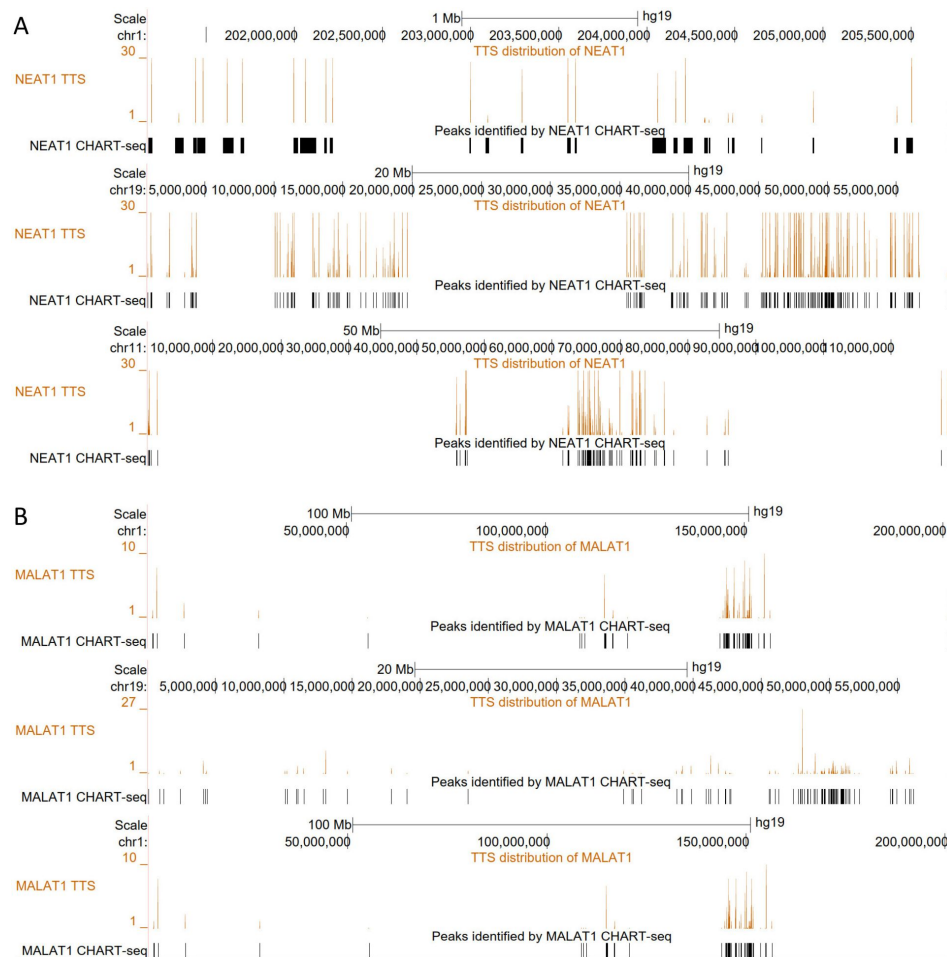
Supplementary Note 3 – Target genes are enriched in specific functions

Strong (binding affinity ≥ 60) and weak (binding affinity < 60) DBSs may indicate well-established and recently-formed gene expression regulation by HS lncRNAs. To examine if genes with strong and weak DBSs are enriched in different biological functions, we used the *g:Profiler* program (with the parameter settings: Organism=Homo sapiens, Ordered query=No, Significance threshold=Benjamini-Hochberg FDR, User threshold=0.05, 50 < terms size < 1000) and the Gene Ontology (GO) database to perform over-representation analysis. An over-representation analysis is used to determine which *a priori* defined gene sets are more present (over-represented) in a subset of “interesting” genes than what would be expected by chance. We used the top 2000 genes with strong DBSs and the bottom 2000 genes with weak DBSs as the input to *g:Profiler* (Supplementary Table 2-4). The two gene sets, with strong and weak DBSs, are enriched in many common GO terms with different significance. These GO terms include “neuron projection development” (GO:0031175) ($2.36E-16$ vs $3.79E-11$), “neuron projection morphogenesis”



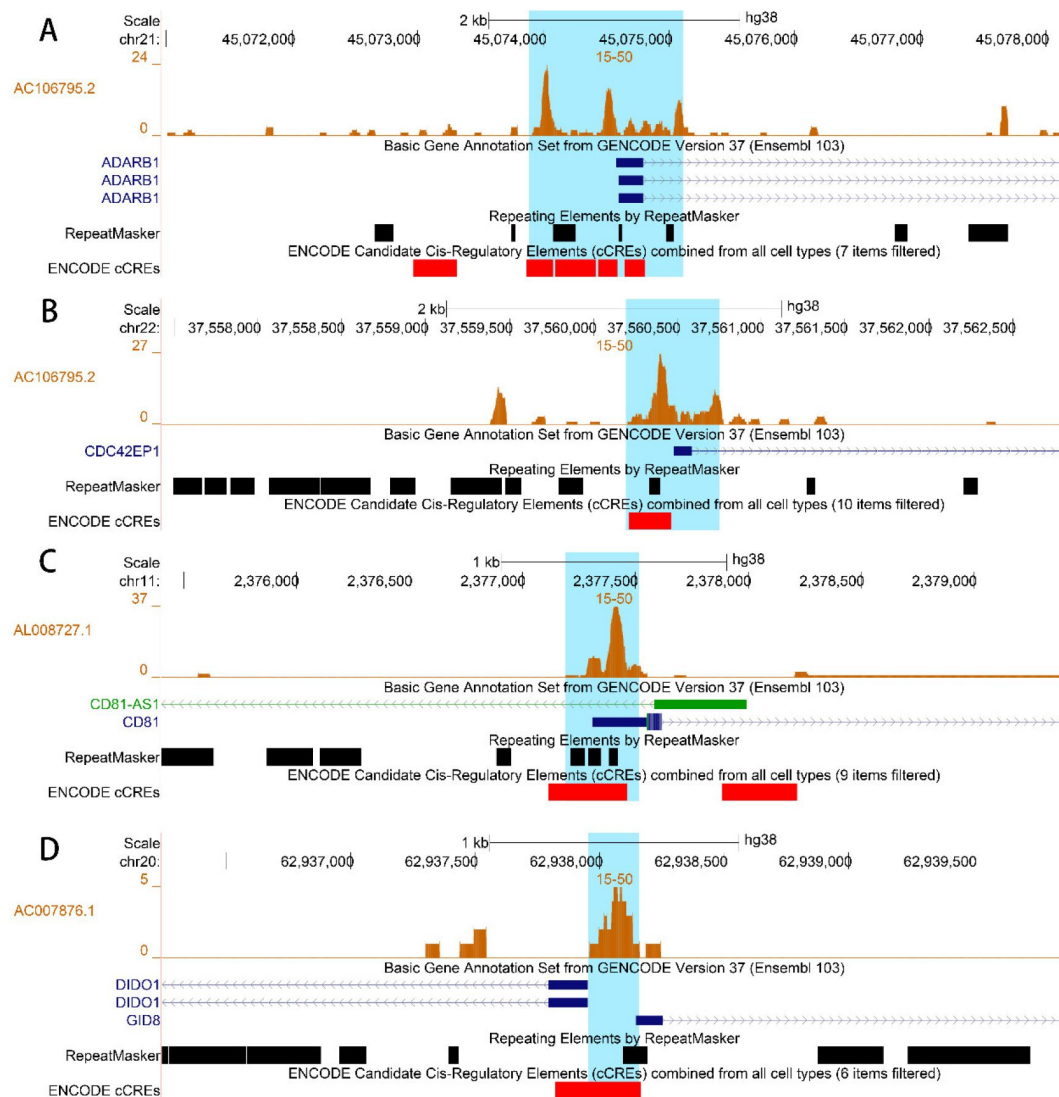
Supplementary Figure 1.

DBSs of the HS lncRNA RP11-423H2.3 in two genomic regions. In the upper and lower panels that display two genomic regions, from top to bottom, are tracks of DBSs (orange peaks), gene annotation, histone modification signals in cell lines, DNA methylation signals in cell lines, H3K4me3 RawSignal, and MRE CpG signals. DBSs overlap very well with DNA methylation and histone modification signals in multiple cell lines.



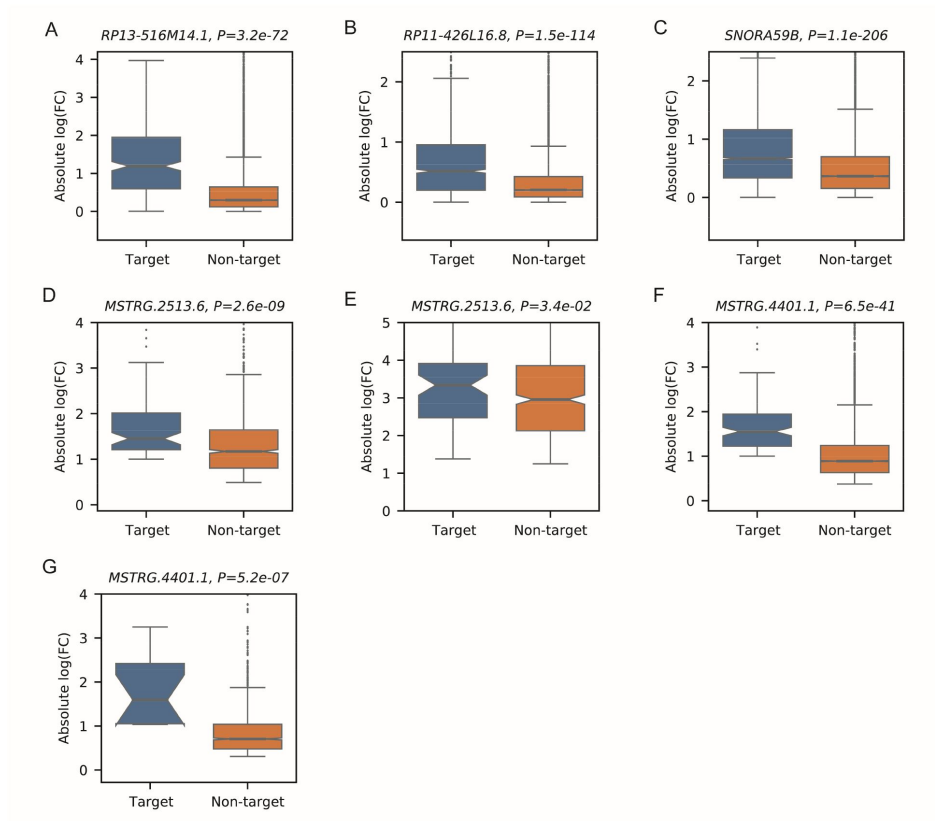
Supplementary Figure 2.

Predicted DBSs and experimentally identified (by CHART-seq) DNA binding sites of NEAT1 and MALAT1 in two cell lines (*West et al., 2014*). DBSs were predicted using the DNA sequences of CHART-seq peaks. 99% and 87% of experimentally identified DNA binding sites of NEAT1 and MALAT1 overlap with predicted DBSs. (A) Predicted DBSs and experimentally identified DNA binding sites of NEAT1 in three genomic regions. (B) Predicted DBSs and experimentally identified DNA binding sites of MALAT1 in three genomic regions.



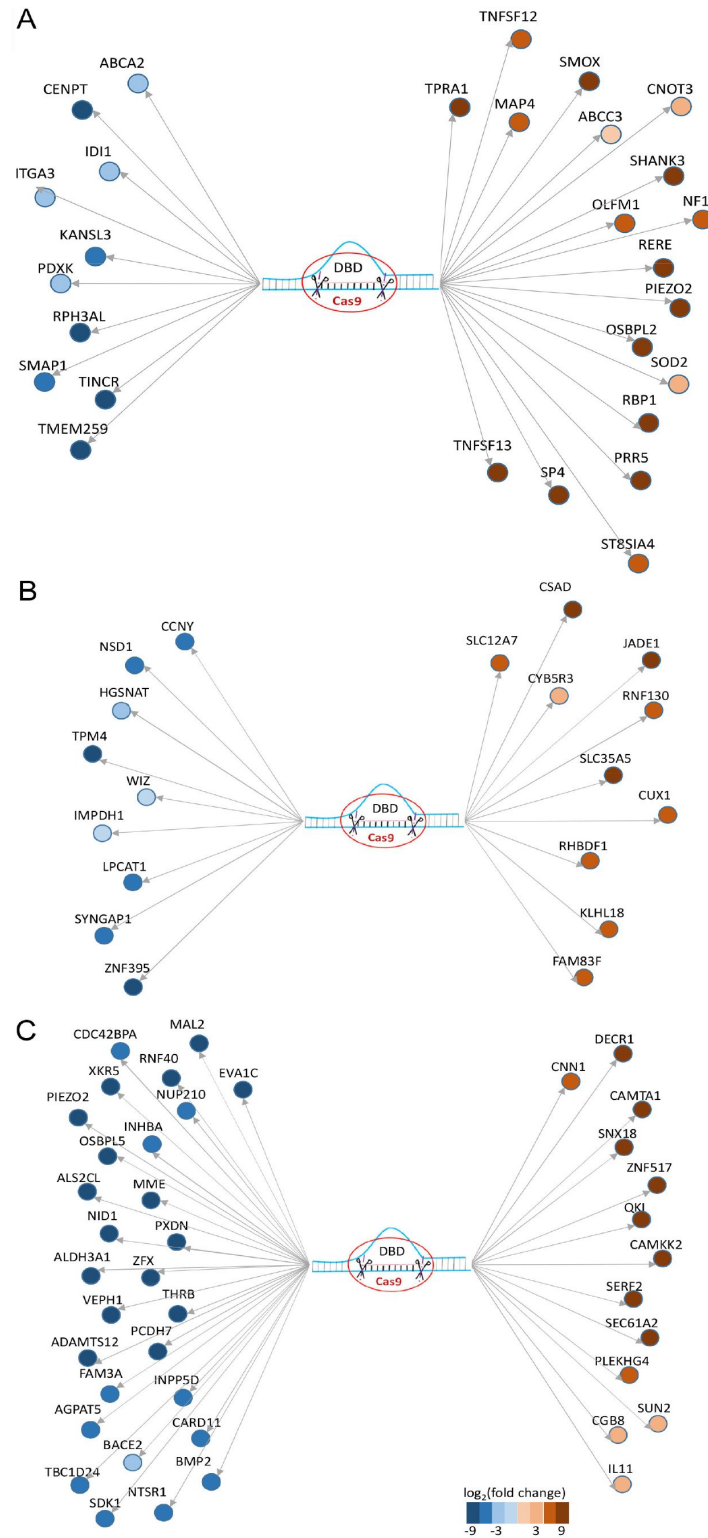
Supplementary Figure 3.

Examples of co-localization of DBSs, TEs, and cCREs in the promoter regions of genes. (A) The DBSs of AC106795.2 in the promoter region of *ADARB1*. (B) The DBSs of AC106795.2 in the promoter region of *CDC42EP1*. (C) The DBS of AL008727.1 in the promoter region of *CD81*. (D) The DBS of AC007876.1 in the promoter region of *DIDO1* and *GID8*.



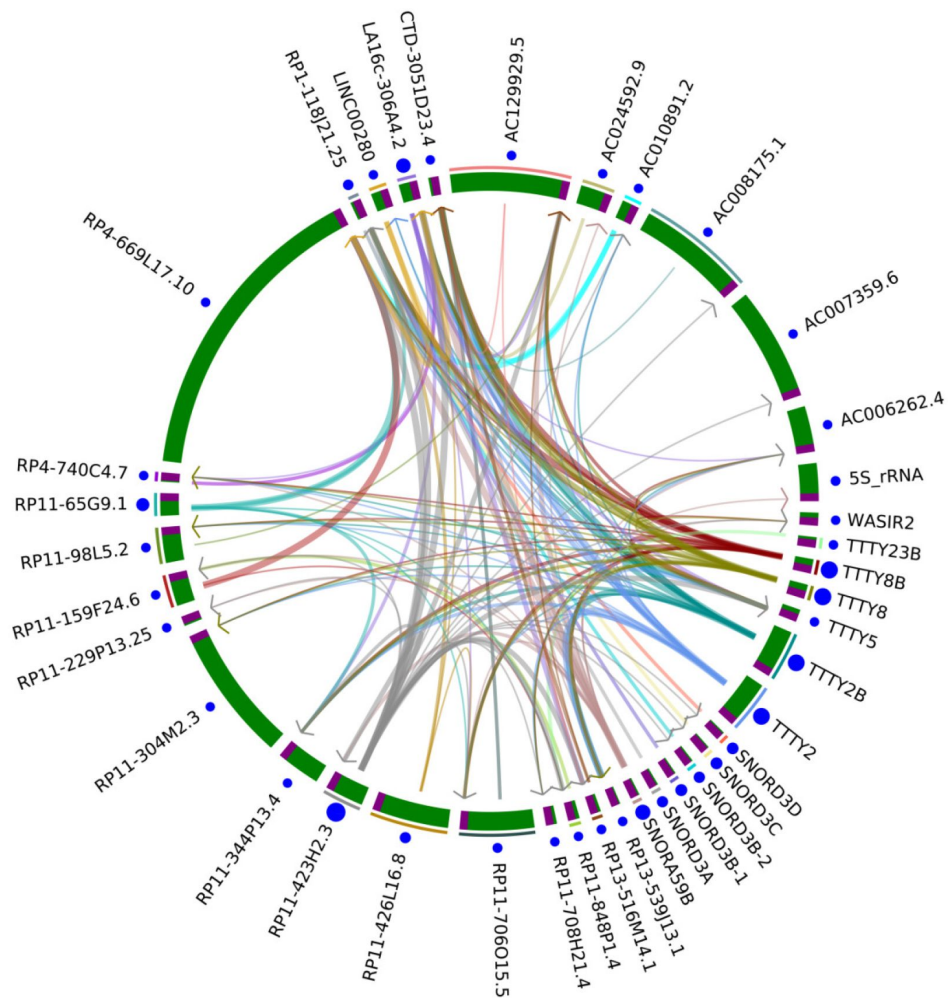
Supplementary Figure 4.

The expression change of target genes was significantly larger than that of non-target genes after DBD knockout. The fold change of gene expression was computed using the edgeR package. The |fold change| distribution of target genes was compared with the |fold change| distribution of non-target genes (one-sided Mann-Whitney test). (A) The knockout of a 157 bp sequence (chr17:80252565-80252721 and contains the DBD of RP13-516M14.1) in the HeLa cell line. (B) The knockout of a 202 bp sequence (chr1:113392603-113392804 and contains the DBD of RP11-426L16.8) in the RKO cell line. (C) The knockout of a 198 bp sequence (chr17:19460524-19460721 and contains the DBD of SNORA59B) in the SK-MES-1 cell line. (D-E) The knockout of the DBD of a wrongly transcribed long noncoding RNA (chr1:156641670-156661464) in the A549 cell line and the HCT116 cell line. (F-G) The knockout of the DBD of a wrongly transcribed long noncoding RNA (chr10:52443915-52455313) in the A549 cell line and the HCT116 cell line. These wrongly transcribed long noncoding RNAs are labeled as “MSTRG” transcripts by the *Stringtie* package.



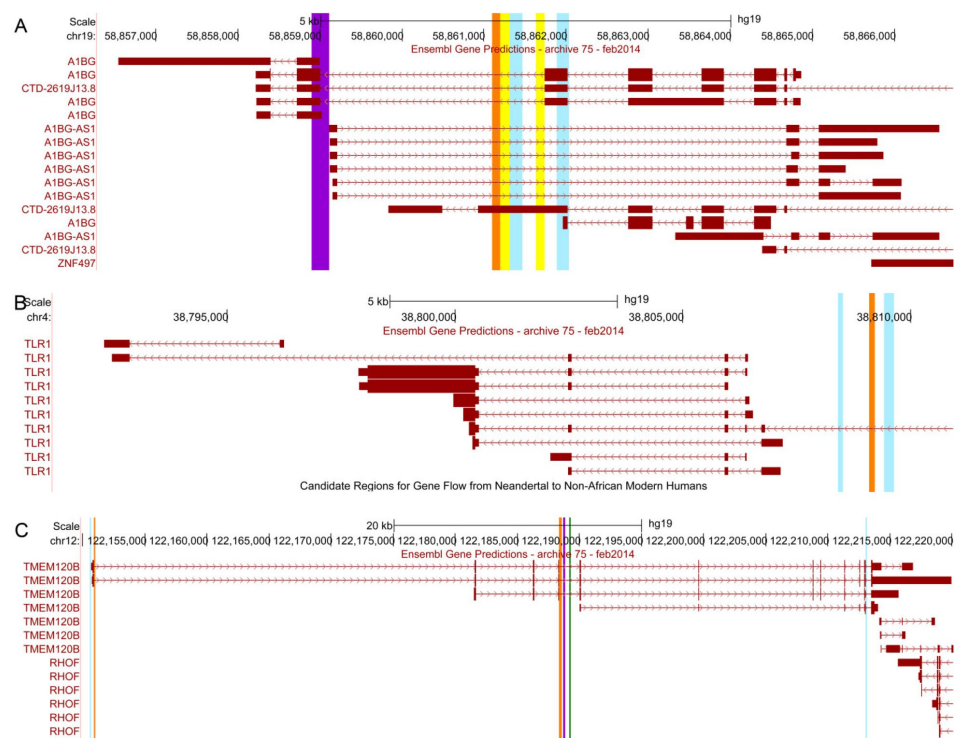
Supplementary Figure 5.

Significant up- and down-regulation ($|\log_2(\text{fold change})| > 1$, $\text{FDR} < 0.1$) of target genes after DBD knockout. (A) RP13-516M14.1. (B) RP11-426L16.8. (C) SNORA59B.



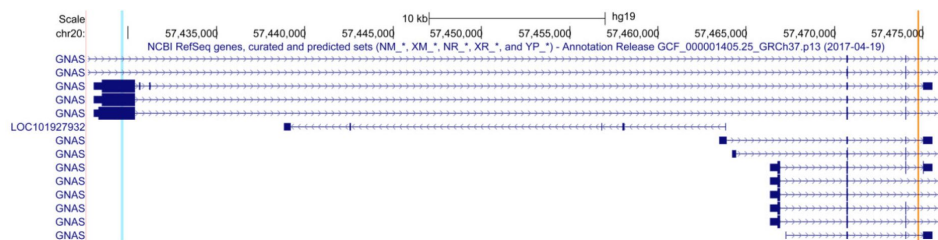
Supplementary Figure 6.

Potential targeting regulation between HS lncRNAs. Brown and green regions in the circle indicate promoter regions and gene body regions. Arrows are from the gene body to promoter regions. The width of the arrows indicates the binding affinity of DBSs, and the sizes of blue dots indicate the number of DBSs of the lncRNA in the genome.



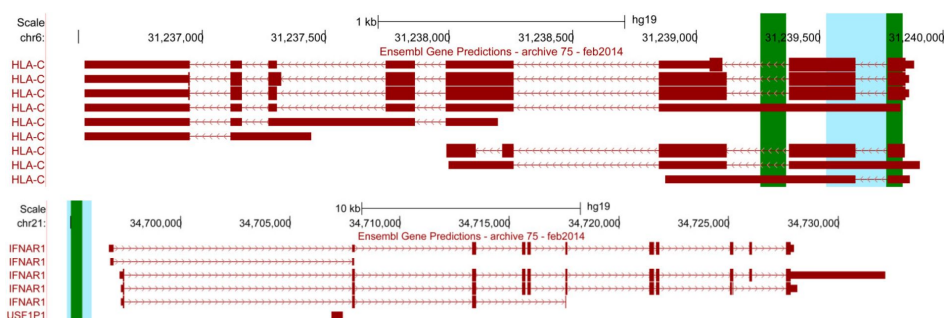
Supplementary Figure 8.

Many genes and transcripts contain DBSs for multiple HS lncRNAs. (A) Left to right: the DBSs of RP11-65G9.1, LA16c-306A4.2, RP13-516M14.1, SNORA59B, RP11-423H2.3, and TTTY8/8B in the *A1BG*. (B) Left to right: the DBSs of TTTY8/8B, RP4-669L17.10, and RP11-423H2.3 in *TLR1*. (C) Left to right: the DBSs of LA16c-306A4.2, RP11-423H2.3, RP11-423H2.3, RP1-118J21.25, RP11-706O15.5, and SNORA59B in *TMEM210B*.



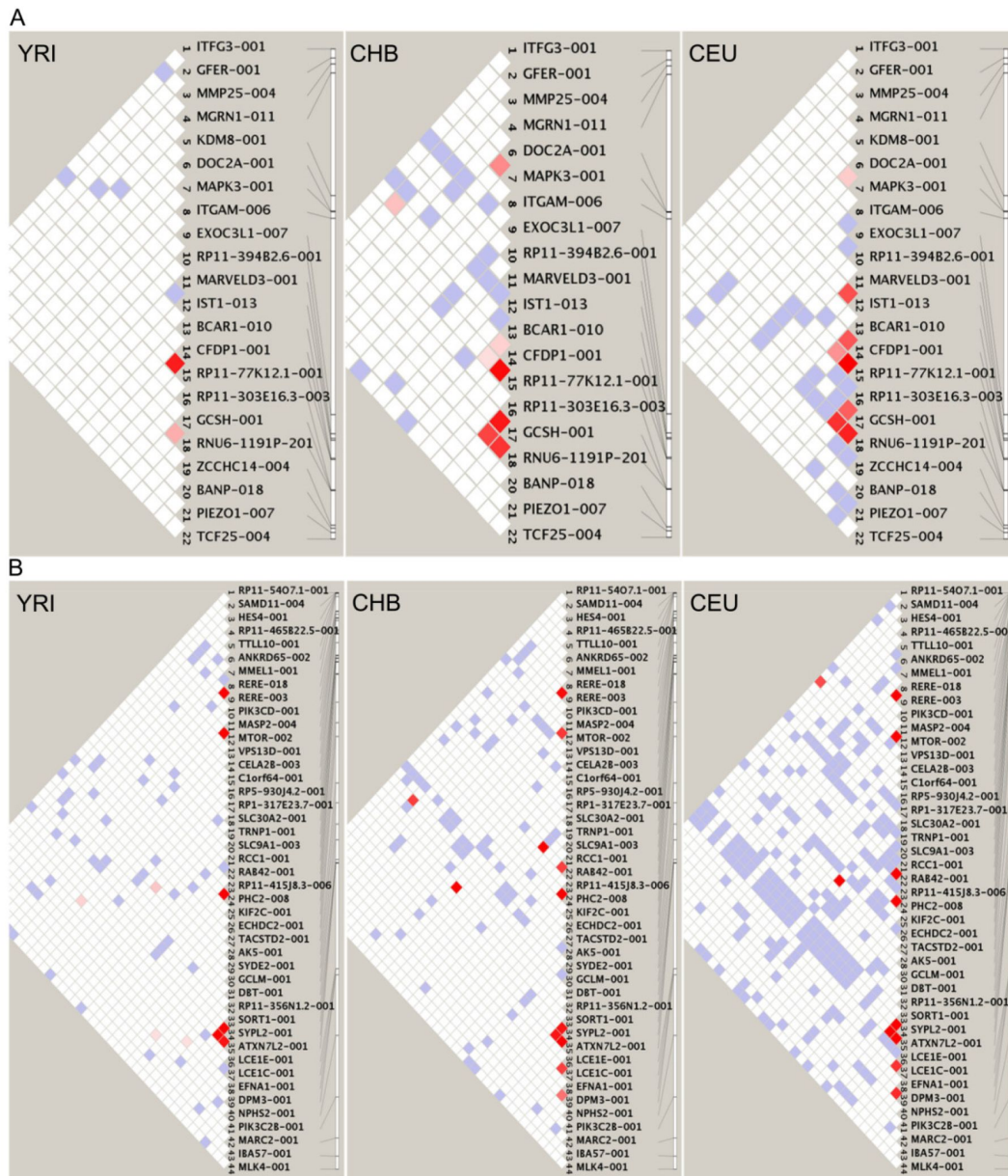
Supplementary Figure 9.

In the *GNAS* region, RP11-423H2.3 has a DBS (indicated by the blue bar) wherein a selection signal was detected in CEU and CHB (Tajima's $D = -0.99/-1.13/1.86$ in CEU/CHB/YRI, integrated $F_{st} = 0.22$), and has another DBS (indicated by the orange bar) wherein a selection signal was detected in YRI (Tajima's $D = 0.25/1.09/-1.17$ in CEU/CHB/YRI, integrated $F_{st} = 0.33$).



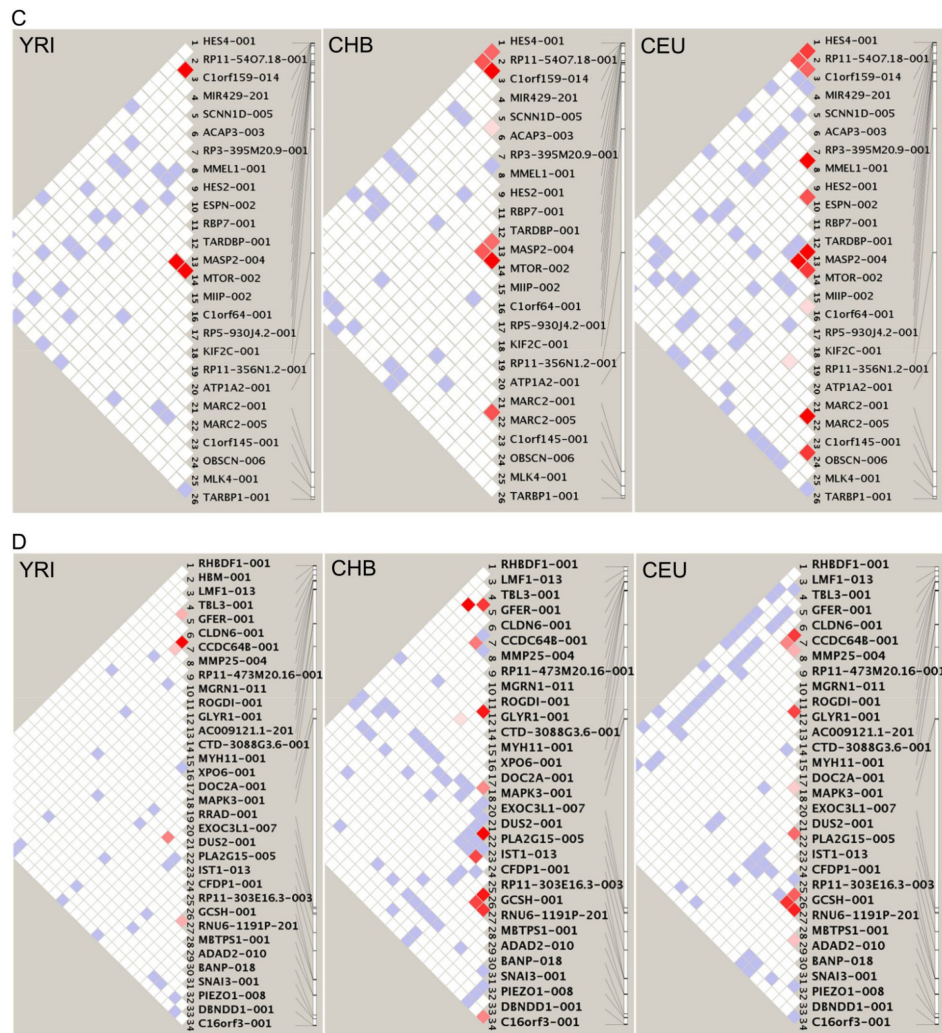
Supplementary Figure 10.

HS lncRNAs on the Y chromosome often have longer DBSs than HS lncRNAs on the autosomes. The top panel shows that the DBS of TTTY2/2B in *HLA-C* (indicated by the blue bar) is longer than the two DBSs of RP11-423H2.3 (indicated by the green bars). The bottom panel shows that the DBS of TTTY8/8B in *IFNAR1* (indicated by the blue bar) is longer than the DBS of LINC00279 (indicated by the green bar).



Supplementary Figure 11.

The LD of the key SNP in DBSs of HS lncRNAs in genes on some chromosomes. (A) The LD of the key SNP in the DBSs of LA16c-306A4.2 in some genes on chromosome 16. (B) The LD of the key SNP in the DBSs of RP11-423H2.3 in some genes on chromosome 1. (C) The LD of the key SNP in the DBSs of SNORA59B in some genes on chromosome 1. (D) The LD of the key SNP in the DBSs of TTTY8B in some genes on chromosome 16.



Supplementary Figure 11. (continued)

(GO:0048812) (8.56E-16 vs 1.64E-09), and “female sex differentiation” (GO:0046660) (2.40E-2 vs 5.84E-04) (**Supplementary Figure 12** [↗](#)). Genes with strong and weak DBSs are also enriched in different GO terms. Especially, genes with weak DBSs are enriched in slightly more GO terms (**Supplementary Figure 13** [↗](#)), including “response to alcohol” (GO:0097305), “negative regulation of locomotion” (GO:0040013), “female gonad development” (GO:0008585), “response to temperature stimulus” (GO:0009266), “cerebral cortex development” (GO:0021987), “DNA methylation or demethylation” (GO:0044728), and “female pregnancy” (GO:0007565) (Supplementary Table 4). These DBSs and target genes reflect the recent adaptive evolution of humans.

Next, we examined DBSs that have most changed sequence distance (“distance” hereafter means the distance per base) in two ways. First, we computed DBS sequence distances from the human ancestor to chimpanzees, archaic humans, and modern humans (hg19) (i.e., from the root node to leaf nodes in the phylogenetic tree); second, we computed DBS sequence distances between modern humans (hg19) and chimpanzees and archaic humans (i.e., between leaf nodes in the phylogenetic tree). In the first type of DBS distances, DBSs in the top 20% (5033) genes have a distance > 0.015 from the human ancestor to modern humans. When the human-chimpanzee ancestral sequence has the ancestral sequence (which means the inferring of ancestral allele is of high confidence), DBS distances from the human ancestor to modern humans are indeed larger than DBS distances from the human ancestor to archaic humans. However, it is unreasonable that more DBSs have distances > 0.015 from the human ancestor to Altai Neanderthals, Denisovans, and even chimpanzees (**Supplementary Figure 14** [↗](#)). We think these wrong distances may be because no archaic human genomes were used to build the human ancestral sequence. According to the file “homo_sapiens_ancestor_GRCh37_e71.README” and the paper “1000 Genomes Project Consortium. A global reference for human genetic variation. Nature 2015”, a high-confidence call in the human ancestor sequence is made when all three sequences - the ancestral (the common ancestor of humans and chimpanzees), the sister (chimpanzees), and the ancestral of the ancestral sequences - agree. We found that only about 64% calls in the human ancestor sequence are high-confidence calls and that DBSs in human ancestral sequences with high-confidence have correct distances to the other five leaf nodes.

To examine whether DBS distances between leaf nodes mainly reflect the genetic changes in the human lineage or in the chimpanzee lineage, we also computed sequence distances for DBSs with the most changed sequences between humans and gorillas. We found that the DBS distances between humans and chimpanzees were significantly correlated with those between humans and gorillas (Spearman’s $\rho = 0.57$, $p = 0.0$) and that these DBSs have a larger distance between humans and gorillas (**Supplementary Figure 15A** [↗](#)). These results suggest that the sequence differences of the most changed DBSs between humans and chimpanzees are driven mainly by the genetic changes occurring in the human lineage but not in the chimpanzee lineage. The same results were observed when the archaic humans were examined (**Supplementary Figure 15B** [↗](#)).

In the second type of distances, DBSs in 429 genes have no counterparts in chimpanzees, the largest distance between chimpanzees and modern humans was 5.383, DBSs in the top 20% (4248) genes in chimpanzees have distances > 0.034, and DBSs having distances > 0.034 in genes in Altai Neanderthals, Denisovans, and Vindija Neanderthals are 1256, 2514, and 134 (Supplementary Table 6). These distances agree with the phylogenetic distances between chimpanzees, archaic humans, and modern humans (**Figure 1AE** [↗](#)). We assigned 10.0 as the distance of DBSs that lack counterparts in chimpanzees. Subsequently, to make the statistical significance of enrichment analysis comparable across chimpanzees, Altai Neanderthals, and Denisovans, we used the top 1256 genes whose DBSs > 0.035 between modern humans and chimpanzees, Altai Neanderthals, and Denisovans as the input to the *g:Profiler* program. Genes in Altai Neanderthals are enriched in more GO terms than genes in chimpanzees, but many of these GO terms are highly related to

human traits (**Figure 2**; Supplementary Table 7). The differences in enriched GO terms in chimpanzees and Altai Neanderthals suggest that HS lncRNA-mediated gene expression regulation has greatly evolved from the human ancestor to Altai Neanderthals.

Finally, we also sorted DBSs by the distances from the human ancestor to the five leaf nodes, identified genes with mostly changed DBSs, and applied over-representation analysis to genes with mostly changed DBSs. It is interesting (and reasonable) to find that genes with largest DBS distances from the human ancestor to Altai Neanderthals and to modern humans are enriched in many common GO terms (e.g., “regulation of anatomical structure size” (GO:0090066), “behavior” (GO:0007610), and “cellular response to xenobiotic stimulus” (GO:0071466)), that the Altai Neanderthal-specific GO terms include “sensory system development” (GO:0048880), “brain development” (GO:0007420), and “adult locomotory behavior” (GO:0008344), and that the modern human-specific GO terms include “female pregnancy” (GO:0007565), “learning” (GO:0007612), and “cognition” (GO:0050890) (Supplementary Table 7). The differences in Neanderthal-specific and modern human-specific GO terms reasonably reflect the different stages of human evolution.

Supplementary Note 4 – Positive selection signals in HS lncRNA genes

We used multiple tests, including XP-CLR (**Chen et al., 2010**), iSAFE (**Akbari et al., 2018**), Tajima’s D (**Tajima, 1989**), the fixation index (Fst) (**Weir and Cockerham, 1984**), and linkage disequilibrium (LD) (**Slatkin, 2008**), to detect positive selection signals in HS lncRNA genes.

First, we used the XP-CLR program to scan the genome regions which contain HS lncRNAs and their 500-kb upstream and downstream sequences. Six pairwise comparisons were applied to the three human populations (CEU-CHB, CEU-YRI, CHB-CEU, CHB-YRI, YRI-CEU, and YRI-CHB). Selective sweeps were detected in the genome regions containing RP11-848P1.4 and RP11-598D14.1 in the CEU-YRI and CHB-YRI comparisons (**Supplementary Figure 16**). Abundant SNPs in these regions have low derived allele frequency (DAF) in YRI, but are nearly fixed in CEU and CHB. Examples (with DAF in YRI, CEU, and CHB) include rs7208589 (DAF = 0.162, 0.990, and 0.995), rs8073226 (DAF = 0.148, 0.990, and 0.995), and rs9915124 (DAF = 0.181, 0.990, and 0.995) in the region containing RP11-848P1.4, and rs4690648 (DAF = 0.185, 0.939, and 0.917), rs11722101 (DAF = 0.269, 0.970, and 0.917), and rs11730933 (DAF = 0.292, 0.970, and 0.917) in the region containing RP11-598D14.1.

Second, we used the iSAFE program to detect favored mutations in HS lncRNA genes in these populations. Because selective sweeps caused by favored mutations may have varied lengths, we ran iSAFE four times with genome regions of 80, 500, 1000, and 2500 kb, which center at the HS lncRNA. Selective sweeps and favored mutations were detected robustly in genome regions containing RP11-598D14.1, RP11-848P1.4, AC006129.1, AC006129.4, CTD-2291D10.1, CTD-2291D10.2, and LA16c-306A4.2 in CEU and CHB, and in the genome region containing CTD-3051D23.4 in CHB (**Supplementary Figure 17**). Mutations with top iSAFE scores were located in the gene body regions of HS lncRNAs, and their DAF values were low in YRI but high in CEU and CHB. In addition, all of the detected favored mutations have high LD (r^2) scores.

Third, we used Tajima’s D and integrated Fst to detect positive selection signals in gene body regions of HS lncRNAs. Positive selection signals were detected in RP11-598D14.1, RP11-848P1.4, RP11-344P13.4, and RP11-426L16.8 in CEU and CHB, in AC129929.5 and RP11-423H2.3 in CEU, and in CTB-151G24.1 in CHB.

Term_name (highest binding affinity)	Term_id	Adjusted_p_value	Term_name (lowest binding affinity)	Term_id	Adjusted_p_value
small GTPase mediated signal transduction	GO:0007264	8.55E-17	neuron projection development	GO:00031175	3.79E-11
neuron projection development	GO:00031175	2.36E-16	cell morphogenesis involved in differentiation	GO:0000904	5.36E-11
cell projection morphogenesis	GO:0048858	5.53E-16	cellular component morphogenesis	GO:0032989	6.81E-11
neuron projection morphogenesis	GO:0048812	8.56E-16	regulation of plasma membrane bounded cell projection	GO:0120035	2.58E-10
plasma membrane bounded cell projection	GO:0120039	8.56E-16	plasma membrane bounded cell projection	GO:0120039	3.44E-10
cell junction organization	GO:0034330	1.96E-15	regulation of anatomical structure morphogenesis	GO:0022603	4.34E-10
cell part morphogenesis	GO:0032990	5.24E-15	cell projection morphogenesis	GO:0048858	4.88E-10
synaptic signaling	GO:0099536	1.38E-14	cell part morphogenesis	GO:0032990	6.25E-10
cellular component morphogenesis	GO:0032989	3.76E-14	regulation of cell projection organization	GO:0031344	1.13E-09
trans-synaptic signaling	GO:0099537	1.88E-13	neuron projection morphogenesis	GO:0048812	1.64E-09
cell morphogenesis involved in differentiation	GO:0000904	3.16E-13	actin filament-based process	GO:0030029	1.65E-09
regulation of small GTPase mediated signal transduction	GO:0051056	4.03E-13	actin cytoskeleton organization	GO:0030036	1.99E-09
chemical synaptic transmission	GO:0007268	4.18E-13	organophosphate metabolic process	GO:0019637	2.05E-09
anterograde trans-synaptic signaling	GO:0098916	4.18E-13	cell morphogenesis involved in neuron differentiation	GO:0048667	2.47E-09
regulation of plasma membrane bounded cell projection	GO:0120035	1.72E-12	regulation of cellular component biogenesis	GO:0044087	8.54E-09
regulation of cell projection organization	GO:0031344	1.74E-12	regulation of neuron projection development	GO:0010975	1.22E-08
cell morphogenesis involved in neuron differentiation	GO:0048667	3.83E-12	cell junction organization	GO:0034330	3.32E-08
dendrite development	GO:0016358	4.71E-11	organophosphate biosynthetic process	GO:0094007	3.82E-08
enzyme-linked receptor protein signaling pathway	GO:0007167	4.71E-11	growth	GO:0040007	3.85E-08
cell surface receptor signaling pathway involved in cell-cell signaling	GO:1905114	4.86E-11	developmental growth	GO:0048589	5.54E-08
actin filament-based process	GO:0030029	7.42E-11	positive regulation of protein modification process	GO:0031401	9.39E-08
regulation of transmembrane transport	GO:0034762	8.88E-11	regulation of cell morphogenesis	GO:0022604	1.08E-07
synapse organization	GO:0050808	1.07E-10	negative regulation of cellular component organization	GO:0051129	1.64E-07
regulation of cellular component biogenesis	GO:0044087	1.17E-10	lipid biosynthetic process	GO:0008610	2.44E-07
metal ion transport	GO:0030001	4.76E-10	positive regulation of transport	GO:0051056	3.52E-07
Ras protein signal transduction	GO:0007265	5.80E-10	regulation of locomotion	GO:0040012	4.21E-07
regulation of ion transport	GO:0043269	7.96E-10	organelle assembly	GO:0070925	4.42E-07
modulation of chemical synaptic transmission	GO:0050804	8.08E-10	regulation of cell migration	GO:0030334	4.90E-07
regulation of trans-synaptic signaling	GO:0099177	8.80E-10	mitotic cell cycle	GO:0000278	6.42E-07
cation transmembrane transport	GO:0098655	1.65E-09	synapse organization	GO:0050808	6.74E-07
regulation of anatomical structure morphogenesis	GO:0022603	1.75E-09	actomyosin structure organization	GO:0031032	1.15E-06
dendrite morphogenesis	GO:0048813	2.21E-09	regulation of cell motility	GO:2000145	1.19E-06
behavior	GO:0007610	2.36E-09	neuron migration	GO:0001764	1.61E-06
transmembrane receptor protein tyrosine kinase	GO:0007169	2.84E-09	positive regulation of cell projection organization	GO:0031346	1.87E-06
circulatory system process	GO:0030013	5.41E-09	smooth muscle cell migration	GO:0014909	2.31E-06
regulation of GTPase activity	GO:0043087	6.17E-09	axon guidance	GO:0007411	2.43E-06
Wnt signaling pathway	GO:0016055	7.92E-09	supramolecular fiber organization	GO:0097435	2.59E-06
regulation of neuron projection development	GO:0010975	8.14E-09	neuron projection guidance	GO:0097485	2.73E-06
cell junction assembly	GO:0034329	8.85E-09	positive regulation of phosphorus metabolic process	GO:0010562	2.83E-06
cell-cell signaling by wnt	GO:0198738	9.83E-09	positive regulation of phosphate metabolic process	GO:0045837	2.93E-06
inorganic ion transmembrane transport	GO:0098660	1.02E-08	mitotic cell cycle process	GO:1903047	3.17E-06
regulation of vesicle-mediated transport	GO:0060627	1.26E-08	muscle cell migration	GO:0014812	3.36E-06
actin cytoskeleton organization	GO:0030036	1.35E-08	cell-substrate adhesion	GO:0031589	3.41E-06
axon development	GO:0061564	1.43E-08	cell projection assembly	GO:0030031	3.49E-06
supramolecular fiber organization	GO:0097435	1.70E-08	non-membrane-bounded organelle assembly	GO:0140694	3.80E-06
inorganic cation transmembrane transport	GO:0098662	2.55E-08	transmembrane receptor protein tyrosine kinase	GO:0007169	4.50E-06
regulation of ion transmembrane transport	GO:0034765	4.34E-08	positive regulation of locomotion	GO:0040017	4.88E-06
blood circulation	GO:0008015	8.44E-08	positive regulation of phosphorylation	GO:0042327	5.13E-06
protein localization to cell periphery	GO:1990778	8.54E-08	head development	GO:0060322	5.78E-06
axonogenesis	GO:0007409	8.84E-08	plasma membrane bounded cell projection assembly	GO:0120031	8.27E-06
actin filament organization	GO:0007615	1.08E-07	nucleobase-containing small molecule metabolic process	GO:0055086	8.56E-06
potassium ion transmembrane transport	GO:0071805	1.64E-07	positive regulation of cell motility	GO:2000147	9.88E-06
peptidyl-serine modification	GO:0018209	2.10E-07	regulation of axonogenesis	GO:0050770	1.06E-05
positive regulation of cell differentiation	GO:0045597	2.30E-07	developmental cell growth	GO:0048588	1.07E-05
regulation of system process	GO:0044057	2.59E-07	positive regulation of cell migration	GO:0030335	1.08E-05
vesicle-mediated transport in synapse	GO:0099003	2.97E-07	organelle localization	GO:0051640	1.26E-05
sodium ion transport	GO:0006814	2.97E-07	microtubule-based process	GO:0007017	1.30E-05
regulation of membrane potential	GO:0042391	3.55E-07	positive regulation of cellular component movement	GO:0051272	1.42E-05
positive regulation of GTPase activity	GO:0043547	5.52E-07	brain development	GO:0007420	1.42E-05
heart development	GO:0007507	5.60E-07	microtubule cytoskeleton organization	GO:0000226	1.42E-05
peptidyl-threonine modification	GO:0018210	5.72E-07	axonogenesis	GO:0007409	1.84E-05
regulation of locomotion	GO:0040012	8.80E-07	ion homeostasis	GO:0050801	2.14E-05
tissue morphogenesis	GO:0048729	8.95E-07	cell growth	GO:0016049	2.14E-05
regulation of cell motility	GO:2000145	9.11E-07	protein deacetylation	GO:0006476	2.43E-05
peptidyl-serine phosphorylation	GO:0018105	1.18E-06	axon development	GO:0061564	2.53E-05
regulation of cell migration	GO:0030334	1.21E-06	chemotaxis	GO:0006935	2.56E-05
sodium ion transmembrane transport	GO:0035725	1.31E-06	regulation of cell size	GO:0008361	2.67E-05
protein localization to plasma membrane	GO:0072659	1.42E-06	nucleotide metabolic process	GO:0009117	2.67E-05
potassium ion transport	GO:0006813	1.86E-06	cell surface receptor signaling pathway involved in cell-cell signaling	GO:1905114	2.77E-05
regulation of cation transmembrane transport	GO:1904062	2.14E-06	taxi	GO:0042330	2.88E-05
growth	GO:0040007	2.35E-06	microtubule cytoskeleton organization involved in mitosis	GO:1902850	3.22E-05
regulation of metal ion transport	GO:0010959	2.53E-06	regulation of vesicle-mediated transport	GO:0060627	3.41E-05
actin filament bundle assembly	GO:0051017	2.69E-06	enzyme-linked receptor protein signaling pathway	GO:0007167	3.51E-05
regulation of growth	GO:0040008	2.85E-06	cell-cell signaling by wnt	GO:0198738	3.54E-05
regulation of Wnt signaling pathway	GO:0030111	3.13E-06	behavior	GO:0007610	3.87E-05
regulation of actin filament organization	GO:0110053	3.50E-06	actin filament organization	GO:0007015	3.87E-05
regulation of cell adhesion	GO:0030155	3.73E-06	sensory organ development	GO:0007423	3.90E-05
regulation of cellular localization	GO:0060341	3.98E-06	neuron projection extension	GO:1990138	4.04E-05

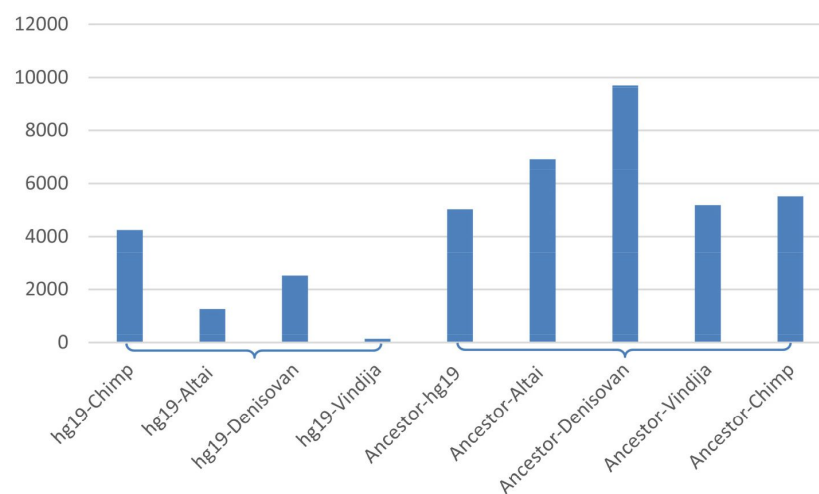
Supplementary Figure 12.

The top 2000 genes with strong binding affinities (left) and the bottom 2000 genes with weak binding affinities (right) are enriched in many common GO terms (marked in pink).

positive regulation of epithelial cell migration	GO:0010634	4.03E-02	vesicle cytoskeletal trafficking	GO:0099518	3.15E-02
establishment of vesicle localization	GO:0051650	4.07E-02	protein complex oligomerization	GO:0051259	3.20E-02
protein localization to extracellular region	GO:0071692	4.19E-02	glycoprotein biosynthetic process	GO:0009101	3.20E-02
positive regulation of small molecule metabolic process	GO:0062013	4.19E-02	negative regulation of catalytic activity	GO:0043086	3.21E-02
negative regulation of growth	GO:0045926	4.22E-02	cell-substrate junction assembly	GO:0007044	3.23E-02
vascular associated smooth muscle cell proliferation	GO:1990874	4.24E-02	regulation of metaphase/anaphase transition of cell cycle	GO:1902099	3.26E-02
nuclear envelope organization	GO:0006998	4.24E-02	chloride transport	GO:0006821	3.31E-02
negative regulation of cell development	GO:0010721	4.35E-02	DNA modification	GO:0006304	3.31E-02
excitatory postsynaptic potential	GO:0006079	4.35E-02	regulation of protein targeting	GO:1903533	3.31E-02
phospholipid biosynthetic process	GO:0008654	4.38E-02	lymphocyte activation involved in immune response	GO:0002285	3.31E-02
negative regulation of smooth muscle cell proliferation	GO:0048662	4.41E-02	inner ear development	GO:0048839	3.31E-02
regulation of cell-cell adhesion	GO:0022407	4.43E-02	ruffle organization	GO:0031529	3.31E-02
regulation of autophagy	GO:0010506	4.46E-02	secretion by cell	GO:0032940	3.33E-02
erythrocyte differentiation	GO:0030218	4.46E-02	positive regulation of chemotaxis	GO:0050921	3.39E-02
regulation of angiogenesis	GO:0045765	4.48E-02	lipid catabolic process	GO:0016042	3.40E-02
regulation of JNK cascade	GO:0046328	4.48E-02	response to ketone	GO:1901654	3.42E-02
myelination	GO:0042552	4.48E-02	mitotic spindle assembly	GO:0090307	3.47E-02
cell chemotaxis	GO:0060326	4.48E-02	regulation of synaptic transmission, glutamatergic	GO:0051966	3.47E-02
receptor internalization	GO:0031623	4.56E-02	response to alcohol	GO:0097305	3.47E-02
apoptotic signaling pathway	GO:0097190	4.56E-02	muscle structure development	GO:0061061	3.47E-02
regulation of tissue remodeling	GO:0034103	4.64E-02	negative regulation of programmed cell death	GO:0043069	3.47E-02
regulation of cell size	GO:0008361	4.66E-02	positive regulation of proteasomal ubiquitin-dependent protein catabolic process	GO:0032436	3.50E-02
leukocyte differentiation	GO:0002521	4.68E-02	regulation of protein kinase B signaling	GO:0051896	3.58E-02
regulation of neuron death	GO:1901214	4.70E-02	purine ribonucleotide biosynthetic process	GO:0009152	3.58E-02
focal adhesion assembly	GO:0048041	4.72E-02	cellular response to environmental stimulus	GO:0104004	3.59E-02
response to metal ion	GO:0019038	4.75E-02	cellular response to abiotic stimulus	GO:0071214	3.59E-02
smooth muscle cell differentiation	GO:0051145	4.75E-02	muscle system process	GO:0003012	3.59E-02
phosphatidylinositol-mediated signaling	GO:0048015	4.84E-02	regulation of mitotic sister chromatid separation	GO:0010965	3.61E-02
cellular carbohydrate metabolic process	GO:0044262	4.84E-02	detection of abiotic stimulus	GO:0009582	3.63E-02
regulation of transmembrane receptor protein	GO:0090092	4.84E-02	regulation of secretion	GO:0051046	3.63E-02
serine/threonine kinase signaling pathway	GO:0010676	4.84E-02	biological process involved in interaction with host	GO:0051701	3.63E-02
positive regulation of cellular carbohydrate metabolic	GO:0051496	4.84E-02	membrane organization	GO:0061024	3.63E-02
positive regulation of stress fiber assembly	GO:0050982	4.84E-02	negative regulation of apoptotic process	GO:0043066	3.66E-02
detection of mechanical stimulus	GO:0048469	4.84E-02	positive regulation of Wnt signaling pathway	GO:0030177	3.69E-02
cell maturation	GO:0045599	4.84E-02	multi-organism reproductive process	GO:0044703	3.69E-02
negative regulation of fat cell differentiation	GO:0097479	4.84E-02	ncRNA transcription	GO:0098781	3.71E-02
synaptic vesicle localization	GO:0006875	4.84E-02	proteoglycan biosynthetic process	GO:0030166	3.71E-02
cellular metal ion homeostasis	GO:0006006	4.85E-02	gamete generation	GO:0007276	3.74E-02
glucose metabolic process	GO:0150116	4.85E-02	transport across blood-brain barrier	GO:0150104	3.74E-02
regulation of cell-substrate junction organization	GO:0062197	4.91E-02	vascular transport	GO:0010232	3.74E-02
cellular response to chemical stress	GO:0008284	4.91E-02	regulation of cell division	GO:0051302	3.76E-02
positive regulation of cell population proliferation	GO:0031330	4.91E-02	TOR signaling	GO:0031929	3.76E-02
negative regulation of cellular catabolic process	GO:0008939	4.91E-02	cellular lipid catabolic process	GO:0044242	3.77E-02
smooth muscle contraction	GO:1901184	4.98E-02	positive regulation of lipid metabolic process	GO:0045834	3.77E-02
regulation of ERBB signaling pathway	GO:0003300	4.98E-02	regulation of mitochondrion organization	GO:0010821	3.77E-02
cardiac muscle hypertrophy			cellular response to inorganic substance	GO:0071241	3.77E-02
			actin filament polymerization	GO:0030041	3.79E-02
			negative regulation of synaptic transmission	GO:0050805	3.80E-02
			glycolipid metabolic process	GO:0006664	3.80E-02
			carboxylic acid catabolic process	GO:0046395	3.80E-02
			regulation of epithelial cell proliferation	GO:0050678	3.83E-02
			response to radiation	GO:0009314	3.85E-02
			protein methylation	GO:0006479	3.86E-02
			protein alkylation	GO:0008213	3.86E-02
			Golgi organization	GO:0007030	3.88E-02
			membrane depolarization	GO:0051899	3.97E-02
			skeletal system morphogenesis	GO:0048705	3.98E-02
			positive chemotaxis	GO:0050918	3.98E-02
			development of primary sexual characteristics	GO:0045137	3.98E-02
			metaphase/anaphase transition of cell cycle	GO:0044784	3.98E-02
			non-motile cilium assembly	GO:1905515	3.98E-02
			bone development	GO:0060348	3.98E-02
			muscle contraction	GO:0006936	4.00E-02
			monovalent inorganic cation homeostasis	GO:0055067	4.00E-02
			chromatin organization	GO:0006325	4.01E-02
			positive regulation of response to external stimulus	GO:0032103	4.06E-02
			negative regulation of transmembrane transport	GO:0034763	4.06E-02
			anatomical structure homeostasis	GO:0060249	4.06E-02
			alpha-amino acid metabolic process	GO:1901605	4.07E-02
			cytokinesis	GO:0000910	4.09E-02
			regulation of regulated secretory pathway	GO:1903305	4.09E-02
			response to calcium ion	GO:0051592	4.09E-02
			regulation of chromosome segregation	GO:0051983	4.09E-02
			protein localization to cytoskeleton	GO:0044380	4.09E-02
			regulation of neurotransmitter transport	GO:0051588	4.09E-02
			liposaccharide metabolic process	GO:1903509	4.09E-02
			nucleotide catabolic process	GO:0009166	4.17E-02
			fatty acid biosynthetic process	GO:0006633	4.26E-02

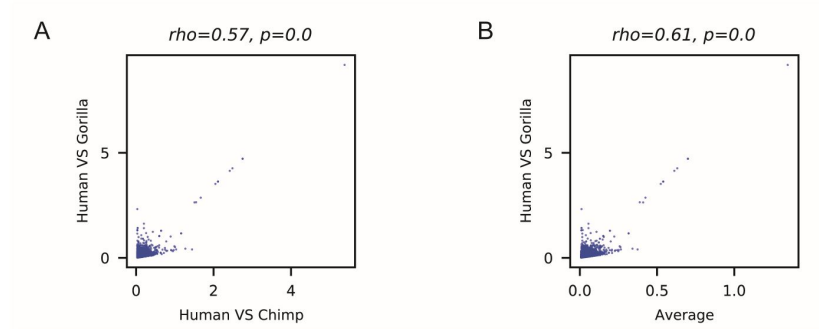
Supplementary Figure 13.

The bottom 2000 genes with weak binding affinities (right) are also enriched in many specific GO terms (marked in yellow) with relatively low significance.



Supplementary Figure 14.

The numbers of DBSs with large distances between modern humans and archaic humans and chimpanzees and from the human ancestor to modern humans and archaic humans and chimpanzees. Left: DBSs in the top 20% (4248) genes in chimpanzees have distances > 0.034 , and DBSs having distances > 0.034 in genes in Altai Neanderthals, Denisovans, and Vindija Neanderthals are 1256, 2514, and 134. Right: DBSs in the top 20% (5033) genes have distances > 0.015 from the human ancestor to modern humans, and DBSs having distances > 0.015 from the human ancestor to Altai Neanderthals, Denisovans, Vindija Neanderthals, and chimpanzees are 6908, 9707, 5189, and 5521.



Supplementary Figure 15.

The most changed DBSs also have large sequence distances between humans and gorillas. (A) Scatter plot showing the sequence distances between humans and chimpanzees and between humans and gorillas. (B) The scatter plot shows the average sequence distances between humans and chimpanzees, the three archaic humans, and between humans and gorillas. The rho and p values were estimated using the Spearman correlation test.

(**Supplementary Figure 18** [↗](#)). Since Tajima's D values were referenced with the genome-wide background, $D < 0$ and $D > 0$ indicate positive (or directional selection) and balancing selection, respectively, instead of population demography dynamics. The F_{st} of each HS lncRNA gene, also referenced with the genome-wide background, was computed for the CEU-YRI, CHB-YRI, and CHB-CEU comparisons. Extreme F_{st} values of SNPs were detected in RP11-598D14.1 and AC129929.5 in the comparisons of CEU-YRI and CHB-YRI. Since F_{st} values were referenced with the genome-wide background, extreme F_{st} values indicate positive selection.

Finally, we applied LD analysis to each HS lncRNA gene. Significantly increased LD was detected in SNPs in AC024592.9, AC129929.5, RP11-157B13.7, RP11-277P12.10, CTD-2142D14.1, and CTD-2291D10.1 in CEU and CHB (**Supplementary Figure 19** [↗](#)). The above results, taken together, indicate that HS lncRNA genes may have undergone more significant adaptive evolution in CEU and CHB than in YRI.

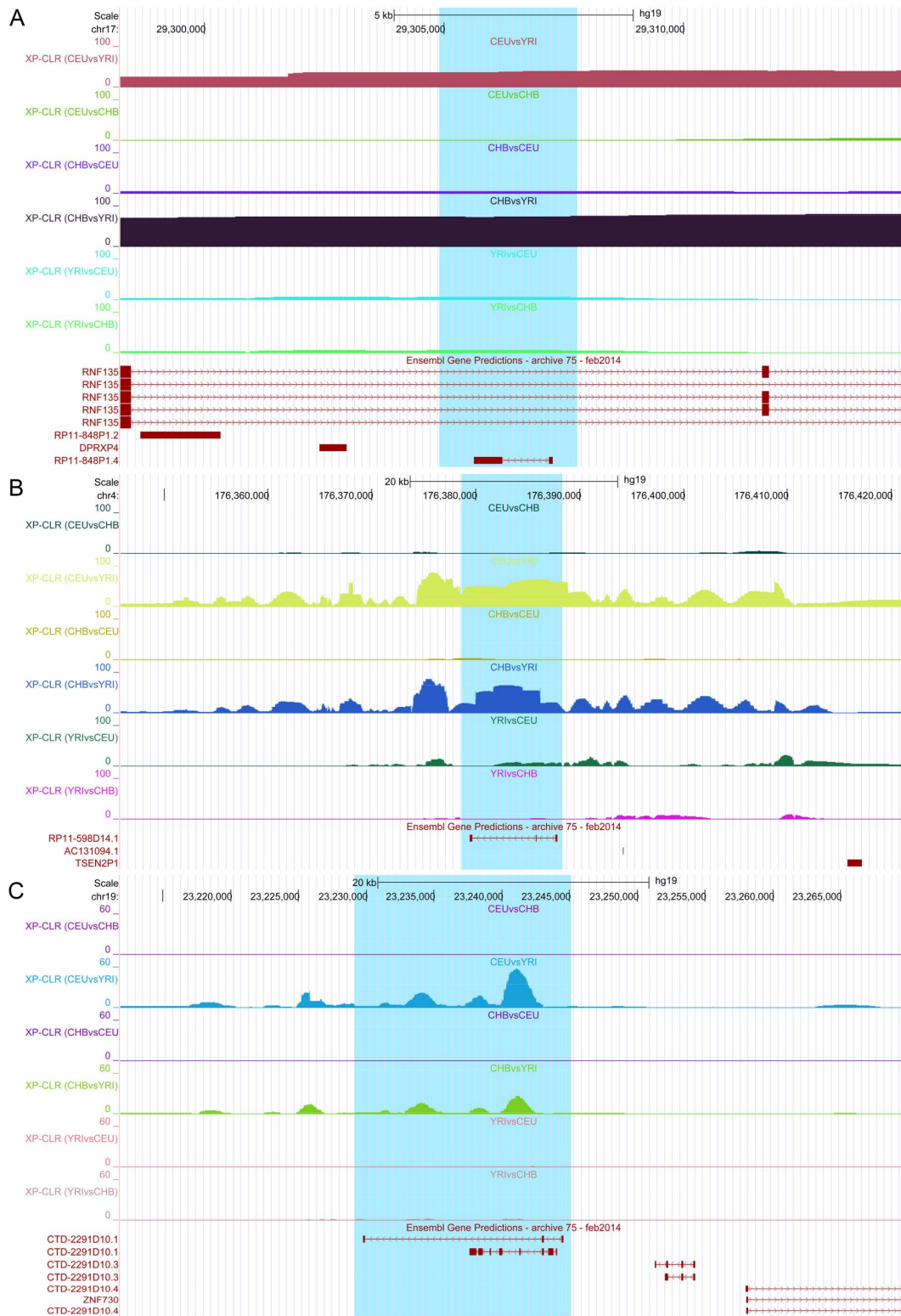
Supplementary Note 5 – Positive selection signals in DBSs

We used the tests mentioned above, including Tajima's D (**Tajima, 1989** [↗](#)), Fay and Wu's H (**Fay and Wu, 2000** [↗](#)), integrated F_{st} (**Weir and Cockerham, 1984** [↗](#)), and LD (**Slatkin, 2008** [↗](#)), to detect positive selection signals in DBSs in each population. These tests unveil positive selection signals in DBSs in specific target genes in specific populations (Supplementary Table 9-11).

First, we calculated Tajima's D, F_{st} , and integrated F_{st} for polymorphic DBSs in CEU, CHB, and YRI. Positive selection signals were detected in DBSs in more genes in CEU and CHB than in YRI (Supplementary Table 10). These genes include the two pigmentation-related genes *MC1R* and *MFSD12*, the two odor reception-related genes *OR6C1* and *TAS1R3*, and the immune-related gene *TLR1*. These target genes may suggest the adaptation of gene expression regulation in CEU and CHB to changes in diet and the environment. In YRI, positive selection signals were detected in DBSs in genes such as *SLCO4A1* which encodes a protein mediating the Na-independent transport of organic anions (e.g., the thyroid hormones T3 and T4). DBSs in different transcripts of *GNAS* (a gene important for genomic imprinting) contain SNPs specifically selected for in different populations (**Supplementary Figure 9** [↗](#)).

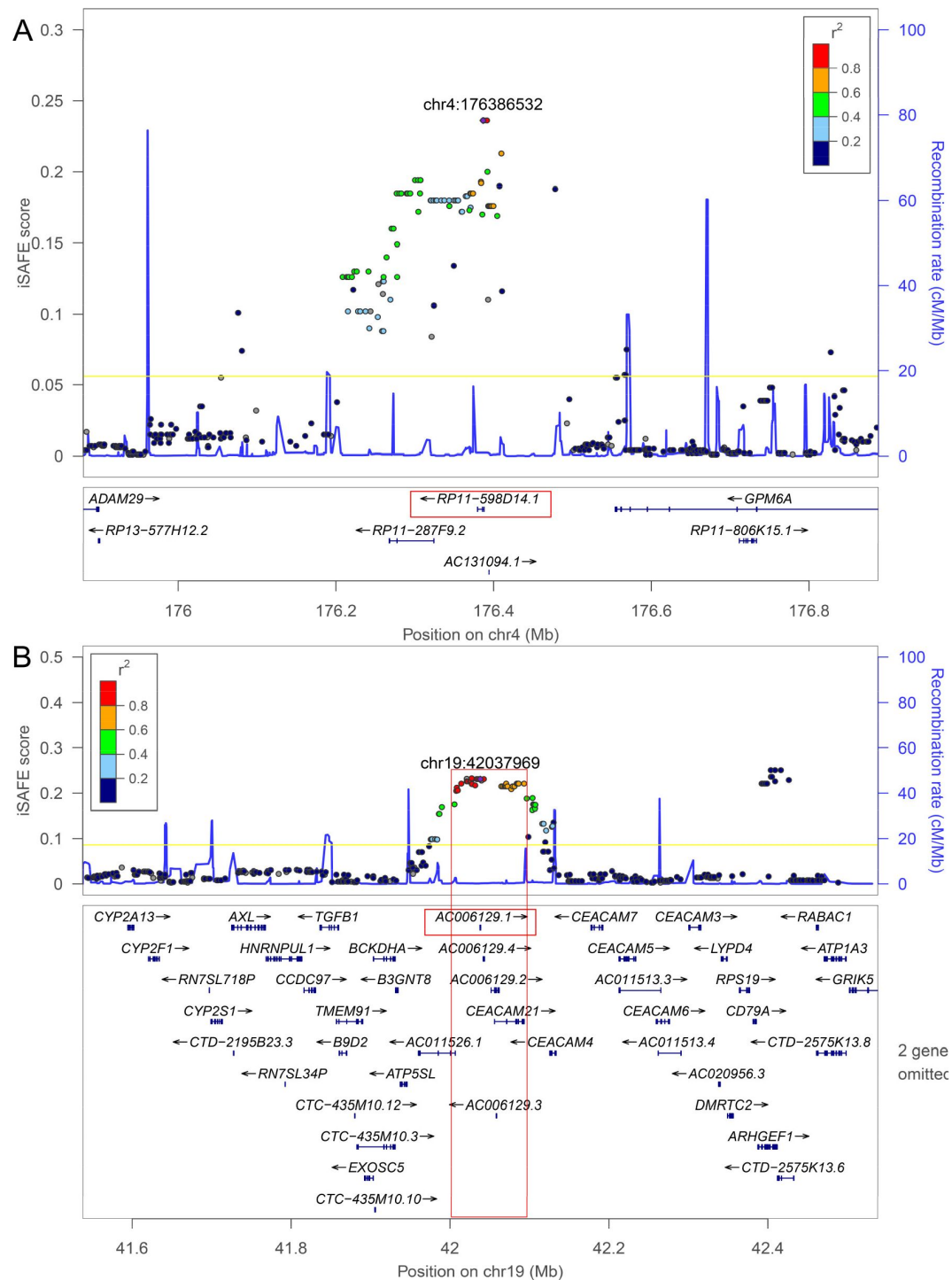
Next, we calculated Fay-Wu's H and integrated F_{st} for each polymorphic DBS in CEU, CHB, and YRI. Strong negative values ($H < -2$), together with significantly large integrated F_{st} (> 0.22), were obtained mainly in DBSs in CEU and CHB (Supplementary Table 11). These genes fall into two classes. The first class includes *PASK*, *CPT1A*, and *EXOC7*, which are important for glucose and lipid metabolism. *PASK* encodes a protein that plays a role in the regulation of insulin gene expression. *CPT1A* encodes a protein that exerts an important role in triglyceride metabolism. *EXOC7* encodes a protein that plays a crucial role in targeting SLC2A4 vesicles to the plasma membrane in response to insulin in adipocytes. The second class includes *COMT*, *TAS1R3*, and *ALMS1*, which are important for neural development. *COMT* is involved in the metabolism of adrenalin and noradrenalin. *TAS1R3* encodes a protein important for recognizing diverse natural and synthetic sweeteners. Mutations in *ALMS1* are associated with Alstrom syndrome.

Third, to examine whether gene expression regulation by HS lncRNAs is coordinated, we examined the LD between SNPs in DBSs of the same HS lncRNA chromosome-wide. When more than one SNP in a DBS has minimal allele frequency (MAF) ≥ 0.05 , we chose the SNP which had the strongest LD (r^2). Despite genetic recombination, LD between DBSs was detected on some chromosomes in CEU and CHB (**Supplementary Figure 11** [↗](#); Supplementary Note 2).



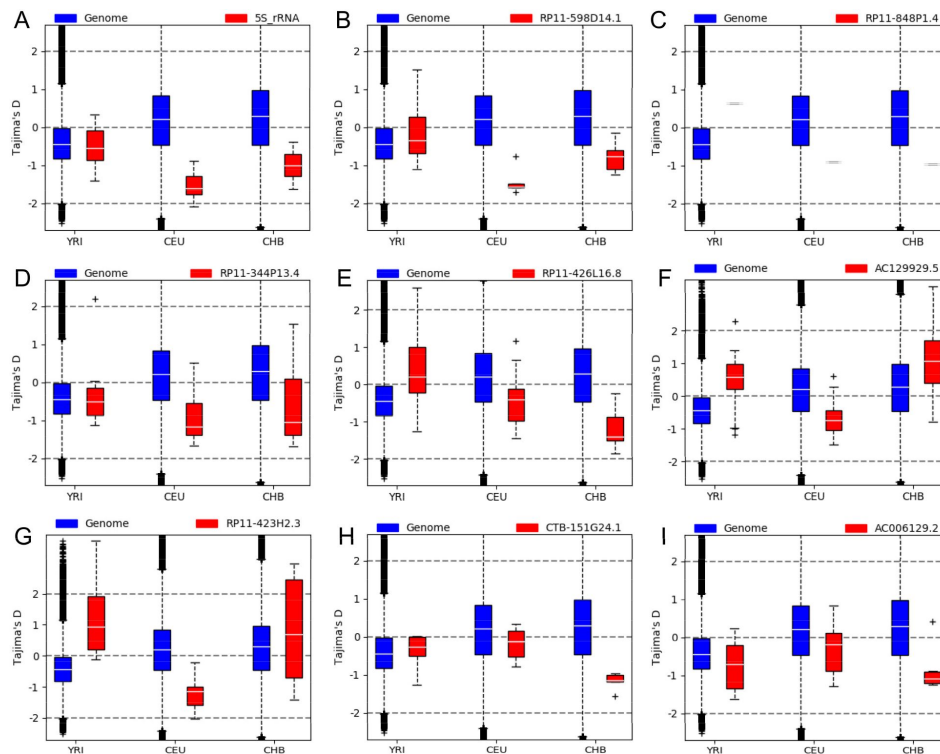
Supplementary Figure 16.

Positive selection signals detected by the XP-CLR program in (A) RP11-848P1.4, (B) RP11-598D14.1, (C) CTD-2291D10.1 in CEU and CHB.



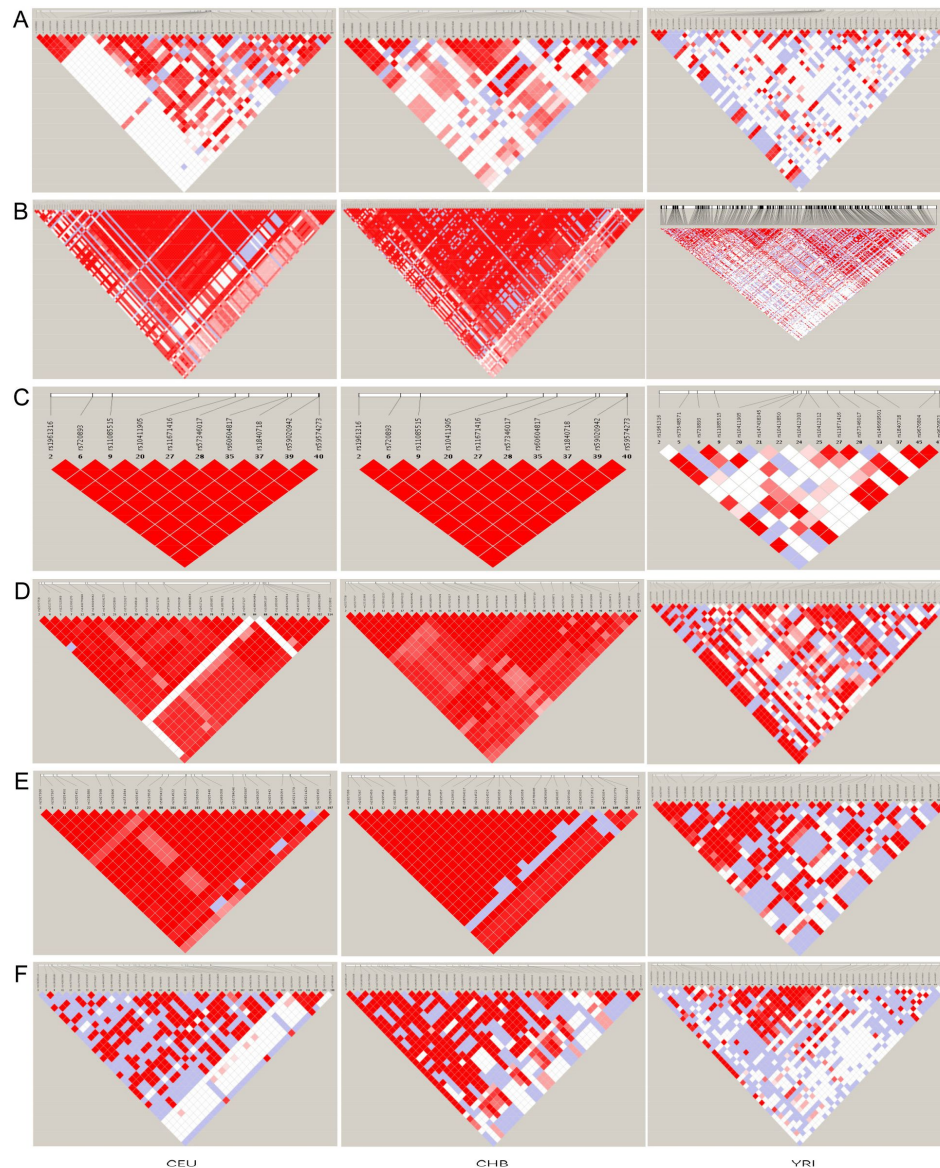
Supplementary Figure 17.

Favored mutations detected by iSAFE. The left and right vertical axes indicate iSAFE scores and recombination rate; the purple diamond marks the top-scored mutation; colors mark LD (r^2) between the top-scored mutation and other ones; the yellow line indicates that the probability that mutations above are neutral is smaller than $p = 1e-6$; the blue curve indicates the position-specific recombination rates. (A) SNPs in the RP11-598D14.1 gene. The top-scored SNP has DAF 0.125/0.960/0.922 in YRI/CEU/CHB. (B) SNPs in the AC006129.1 gene. The top-scored SNP has DAF 0.134/0.717/0.587 in YRI/CEU/CHB.



Supplementary Figure 18.

HS lncRNA genes with significantly changed Tajima's D in CEU, CHB, and YRI. Negative and positive Tajima's D scores, which are significantly smaller or larger than the genome-wide background in a population, indicate the signature of positive selection or balancing selection, respectively, in the population.



Supplementary Figure 19.

The LD of SNPs in HS lncRNA genes in CEU, CHB, and YRI. The red color indicates high LD values. These panels show that LD between SNPs in CEU and CHB in these lncRNA genes is stronger than LD between SNPs in YRI. (A) AC024592.9. (B) AC129929.5. (C) RP11-157B13.7. (D) RP11-277P12.10. (E) CTD-2291D10.1. (F) CTD-2142D14.1.

Fourth, we computed the distances between DBSs in modern humans and their counterparts in archaic humans and compared the distances with the distances between annotated promoters and their counterparts in archaic humans. A considerable portion of DBSs (note that DBSs are within promoter regions) have larger distances than promoters between modern and archaic humans (**Supplementary Figure 20** [↗](#)). We also computed the frequency distribution of SNPs (> 0.05) in DBSs and found that SNPs are enriched with low and high frequencies (**Supplementary Figure 21** [↗](#)), which is an indicator of positive selection.

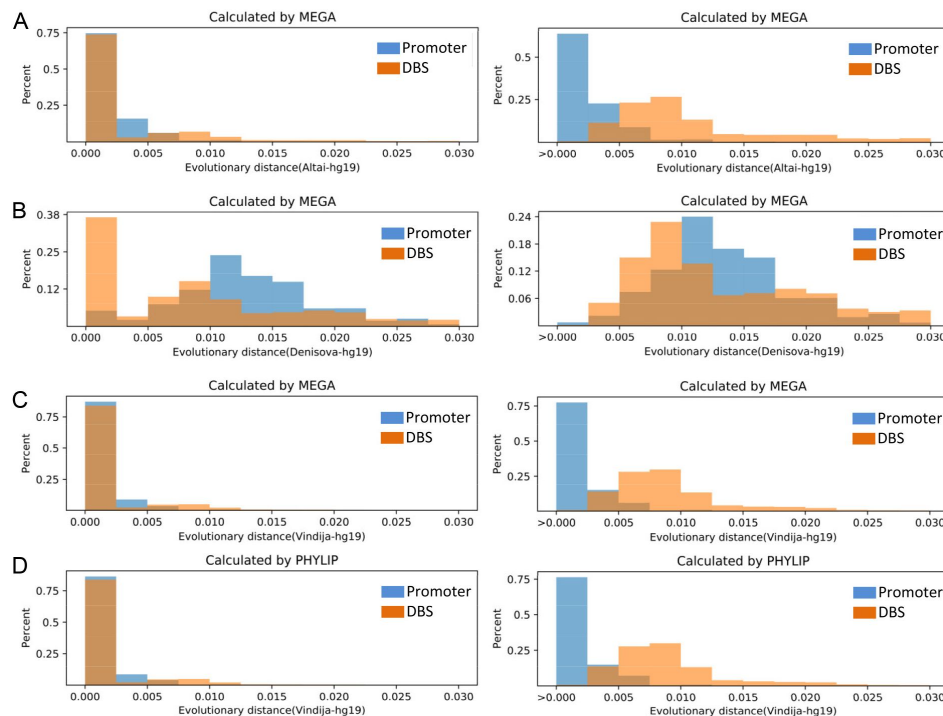
Finally, we analyzed two experimental datasets. It was reported that the expression of a set of genes in T cell activation in response to pathogens shows significant variations in 348 healthy European, African, and Asian individuals (**Ye et al., 2014** [↗](#)). To examine whether HS lncRNAs contribute to these variations, we examined whether DBSs in these genes have different binding affinities in different populations. We found that DBSs in *IFITM3*, *IL2RA*, *IL17F*, *MXRA7*, *CCL22*, and *FADS2* contain SNPs with biased frequencies in CEU, CHB, and YRI. A typical example is the DBSs in *FADS2* which is expressed differentially in Europeans and Africans. The two DBSs in *FADS2-003* and *FADS2-010*, respectively, contain four SNPs - rs71046746 has DAF 0.96/0.99/0.88 in CEU/CHB/YRI, but the other three unannotated SNPs have DAFs 0.24/0.22/0.01, 0.24/0.22/0.04, and 0.24/0.22/0.02 in CEU/CHB/YRI.

Another study examined genome-wide patterns of selection in 230 West Eurasians who lived between 6500 and 300 BC and identified selection signals in multiple genes which are associated with diet, pigmentation, and immunity (**Mathieson et al., 2015** [↗](#)). We detected signatures of positive selection in DBSs in *LCT*, *TLR1*, *TLR6*, *TLR10*, *SLC45A2*, *SLC22A4*, *MHC*, *ZKSCAN3*, *FADS1*, *FADS2*, *DHCR7*, *GRM5*, *ATXN2*, and *HERC2*. Reliable population-specific selection signals were identified in the DBS of RP11-423H2.3 in *HERC2* (the Tajima's D in CEU/CHB/YRI are -0.19/1.82/-1.12 and the integrated Fst is 0.27), in the DBSs of RP11-423H2.3 in *TLR1* and *TLR6* (the Tajima's D in CEU/CHB/YRI are -1.26/-1.2/1.38 and the integrated Fst is 0.24), and in the DBS of SNORA59B in *TLR1* and *TLR6* (the Tajima's D in CEU/CHB/YRI are -1.73/-0.9/1.76 and the integrated Fst is 0.24). Taken together, the above analyses suggest that population-specific selection signals in DBSs may help explain the phenotypic and physiological differences between human populations.

Supplementary Note 6 – Considerable SNPs in DBSs have cis-effect on gene expression in specific tissues and populations

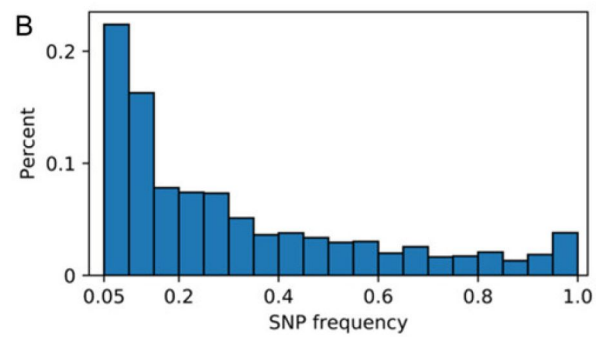
To find supporting evidence for DBS influencing gene expression in tissues and organs, we first examined whether SNPs in DBSs have a cis-effect on gene expression using the data of the Genotype-Tissue Expression (GTEx) project (**GTEx Consortium, 2017** [↗](#)). The expression of some genes in specific tissues is of interest. For example, of the 21 SNPs which are eQTLs exclusively in the GTEx tissue Thyroid, 14 have high DAF in YRI (**Supplementary Figure 22** [↗](#)). Correspondingly, in Africans, positive selection signals were found in genes involved in energy metabolism (**Fan et al., 2019** [↗](#)). Second, we computed the eQTL density for each DBS and each Ensembl-annotated promoter and compared eQTL density of the two kinds of sequences. We found that eQTLs were more enriched in DBSs than in promoters (one-sided Mann-Whitney test, $p = 0.0$) (**Supplementary Figure 23** [↗](#)). Third, for DBS harboring eQTLs in specific tissues, 94% of the corresponding HS lncRNA-target transcript pairs exhibit expression correlation ($|\text{Spearman's } \rho| > 0.3$ and $\text{FDR} < 0.05$) in the eQTLs' tissues.

Finally, we examined how many SNPs DBSs are DNA methylation QTL (mQTL) and histone modification QTL (haQTL). One study analyzed DNA methylation data from three populations (10 Caucasians, 10 African Americans, and 10 Japanese) and identified *RGS6*, *CLEC2L*, *ABCF1*,



Supplementary Figure 20.

The distributions of sequence distances between DBSs in modern and archaic humans and between Ensembl-annotated promoters in modern and archaic humans, with the right-hand panels illustrating the distributions of distance > 0.005. (A) The distances between modern humans and Altai Neanderthals. (B) The distances between modern humans and Denisovans. (C) The distances between modern humans and Vindija Neanderthals. It is most prominent in panel B that a fraction of DBSs has larger distances than promoters, agreeing with the finding of less gene flow from Denisovans to modern humans (Meyer et al., 2012). However, it is least prominent in panel C, agreeing that the evolutionary distance between Vindija Neanderthals and modern humans is short (Figure 1A).



Supplementary Figure 21.

The distribution of SNPs frequencies (MAF > 0.05) in DBSs.

MIR3678, and *EP400* as having differential methylation in African Americans compared with Japanese and Caucasians (Giri et al., 2017). Based on this study, we found that DBSs of several HS lncRNAs in some transcripts of *RGS6*, *CLEC2L*, and *EP400* (also *EP400NL*) had significantly different Tajima's D and integrated Fst in YRI. Another study applied QTL analysis to a multi-omics dataset and identified a set of SNPs significantly associated with levels of gene expression, DNA methylation, and histone modification in the human dorsolateral prefrontal cortex (Ng et al., 2017). Ng et al. called SNPs with cis-effect (including eQTL, mQTLs, and haQTLs) xQTL and found that xQTL SNPs were enriched close to transcription start sites. We re-examined these xQTL SNPs and identified 4735 in the DBSs of HS lncRNAs (Supplementary Table 13). Notably, 4319 out of the 4735 xQTLs were meQTL SNPs, many of which had biased frequencies in CEU, CHB, and YRI.

Supplementary Note 7 – HS lncRNAs critically regulate genes important for brain development

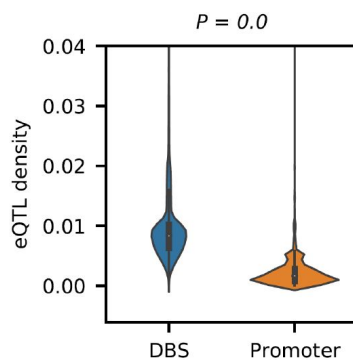
The enlarged brain is one of the most prominent features of modern humans. We analyzed multiple datasets of experimental studies to examine the impact of HS lncRNAs on brain development. First, we analyzed two datasets of epigenetic studies. A comparative H3K27ac and H3K4me2 profiling of human, macaque, and mouse corticogenesis revealed that strongly concordant epigenetic gains are enriched in promoters of 301 genes which are active during human corticogenesis (Reilly et al., 2015). Of these genes (257 are annotated in hg38), 84% have DBSs for at least one HS lncRNA and 56% have DBSs for ≥ 5 HS lncRNAs. Thus, these 301 genes are enriched significantly for regulation by HS lncRNAs compared with genes without DBSs of HS lncRNAs ($p = 1.32e-26$ for the situation of 56% genes and $p = 1.21e-21$ for the situation of 84% genes, two-sided Fisher's exact test). Another study examined gene expression and H3K27ac modification in eight brain regions in humans and four other primates, and revealed 1851 genes (1687 genes annotated in hg38) with human-specific transcriptome differences and 240 genes with chimpanzee-specific transcriptome differences in at least one brain region (Xu et al., 2018). Of these genes, 73% have DBSs for at least one HS lncRNA and 46% have DBSs for ≥ 5 HS lncRNA. These data again indicate that genes related to brain development are enriched significantly in HS lncRNA targets ($p = 1.1e-75$ for the situation of 46% genes and $p = 1.2e-56$ for the situation of 73% genes, two-sided Fisher's exact test).

Second, we analyzed the data from two studies conducted by the PsychENCODE consortium. The data of one study cover 16 brain regions and 9 time windows and include spatiotemporally differentially expressed genes, spatiotemporally differentially methylated sites, spatiotemporal variations of H3K27ac enrichment, and cell type marker genes (Li et al., 2018). Of the 66 HS lncRNAs, 65 were expressed in all of the 16 brain regions and in the 9 time windows, and 7 HS lncRNAs (RP11-423H2.3, RP11-706O15.5, RP13-539J13.1, SNORD3B-1, SNORD3B-2, RP4-740C4.7, and RP13-516M14.1) exhibited spatial or temporal expression differences. To assess the contribution of these 7 HS lncRNAs to the spatiotemporal changes of gene expression, we mapped the spatiotemporally differentially methylated sites to the promoters of annotated transcripts and identified 109 transcripts which had these sites in their promoters. We then predicted the DBSs of the 7 HS lncRNAs in the promoter regions of the 109 transcripts and found that 56 transcripts had DBSs for at least one of the 7 HS lncRNAs. In addition, 61% of 770 cell type-specific marker genes had at least one DBS of the 7 HS lncRNAs. These results suggest that HS lncRNAs regulate spatiotemporal gene expression and determine cell fate during brain development, probably by regulating DNA methylation in promoter regions. The other PsychENCODE study examined the spatiotemporal transcriptomic divergence across human and macaque brain development (Zhu et al., 2018). We examined the potential relationships between the HS lncRNAs and the 8951 genes showing differential expression between human and macaque brain during brain development. We identified DBSs of at least one HS lncRNAs in the promoter regions of 72% of differentially expressed genes in the human brain. Moreover, DBSs of at least 5 HS lncRNAs were identified in

SNP ID	CEU- frequency	CHB- frequency	YRI- frequency
rs75508216	0.01	0.05	0.1
rs114086993	0.01	0.05	0.1
rs201187971	0.01	0	0.1
rs73677017	0	0	0.14
rs11944829	0	0	0.14
rs114884549	0	0	0.15
rs77133472	0	0	0.15
rs115688283	0	0	0.17
rs113131895	0	0	0.17
rs142522981	0	0.02	0.19
rs112731299	0	0	0.2
rs4565803	0.01	0	0.24
rs4604779	0.01	0	0.24
rs76612433	0.02	0.19	0.24

Supplementary Figure 22.

The 14 SNPs have high DAF in YRI and are eQTLs exclusively in the GTEx tissue Thyroid.



Supplementary Figure 23.

DBSs have significantly higher eQTL density than promoters. DBSs and promoters harboring at least one eQTL were used to compute eQTL density and make the comparison. A one-sided Mann-Whitney test was used to compute the p value.

the promoter regions of 44% of differentially expressed genes. Compared with genes without DBS for HS lncRNAs, these differentially expressed genes are significantly enriched for regulation by HS lncRNAs ($p = 0$, Chi-square test). Of the 7 transcription factor genes differentially expressed between humans and macaques, four had DBSs of HS lncRNAs. These results support that gene expression in the human brain is highly regulated by HS lncRNAs.

Third, three recent studies identified several genes, including *NOTCH2NL* and *Aspm*, which are critical for regulating cortical expansion in the human brain (Florio et al., 2018; Johnson et al., 2018; Suzuki et al., 2018). *NOTCH2NL* is highly expressed in radial glia and activates Notch signaling to promote the clonal expansion of human cortical progenitors (Florio et al., 2018; Suzuki et al., 2018). *Aspm* is expressed in mice but shows no contribution to mouse corticogenesis. Nevertheless, an *Aspm* knockout greatly influenced corticogenesis in ferrets (Johnson et al., 2018). We examined whether HS lncRNAs potentially regulate these genes. Of the 40 protein-coding genes reported by the three studies, 14 have DBSs of ≥ 5 HS lncRNAs and 29 have DBSs of ≥ 1 HS lncRNAs in their promoter regions (Supplementary Table 14). Compared with the background situation (22562 protein-coding genes' promoter regions contain, but 20944 protein-coding genes' promoter regions do not contain, DBSs of HS lncRNAs), the 40 protein-coding genes involved in cortical expansion are highly enriched for regulation by HS lncRNAs ($p < 0.01$, two-sided Fisher's exact test).

Recently, brain organoids have emerged as an important approach to studying primate neural development in vitro. By establishing and comparing cerebral organoids between humans, chimpanzees, and macaques, Pollen et al. identified 261 human-specific gene expression changes (Pollen et al., 2019). In another study, Agolia et al. generated a panel of tetraploid human-chimpanzee hybrid iPS cells (i.e. hybrid induced pluripotent stem cells) by fusing human and chimpanzee iPS cells, and differentiated the hybrid iPS cells into hybrid cortical spheroids (Agolia et al., 2021). By allele-specific expression (ASE) analysis, Agolia et al. identified thousands of genes with divergent expression between humans and chimpanzees at at least one time point. We obtained the 261 genes from the first study and the 1102 genes with $|\log FC| > 1$ and adjusted $p < 0.05$ from the second study. Compared with the background situation mentioned above, the two sets of genes were significantly enriched with DBSs of HS lncRNAs ($p = 1.2\text{e-}16$ and $3.4\text{e-}74$, respectively).

References

1. 1000 Genomes Project Consortium, Abecasis G.R., Auton A., Brooks L.D., DePristo M.A., Durbin R.M., Handsaker R.E., Kang H.M., Marth G.T., McVean G.A. (2012) **An integrated map of genetic variation from 1,092 human genomes** *Nature* **491**:56–65
2. Almakarem Abu, Petrov A.S., Stombaugh A.I., Zirbel J., and Leontis C.L. (2012) **Comprehensive survey and geometric classification of base triples in RNA structures** *Nucleic Acids Res* **40**:1407–1423
3. Agoglia R.M., Sun D., Birey F., Yoon S.J., Miura Y., Sabatini K., Pasca S.P., Fraser H.B (2021) **Primate cell fusion disentangles gene regulatory divergence in neurodevelopment** *Nature* **592**:421–427
4. Akbari A., Vitti J.J., Iranmehr A., Bakhtiari M., Sabeti P.C., Mirarab S., Bafna V (2018) **Identifying the favored mutation in a positive selective sweep** *Nat Methods* **15**:279–282
5. Albalat R., Canestro C (2016) **Evolution by gene loss** *Nat Rev Genet* **17**:379–391
6. Anton S.C., Potts R., Aiello L.C (2014) **Human evolution. Evolution of early Homo: an integrated biological perspective** *Science* **345**
7. Barrett J.C., Fry B., Maller J., Daly M.J (2005) **Haploview: analysis and visualization of LD and haplotype maps** *Bioinformatics* **21**:263–265
8. Breschi A., Gingeras T.R., Guigo R (2017) **Comparative transcriptomics in human and mouse** *Nat Rev Genet* **18**:425–440
9. Chen H., Patterson N., Reich D (2010) **Population differentiation as a test for selective sweeps** *Genome Res* **20**:393–402
10. Chimpanzee Sequencing and Analysis Consortium (2005) **Initial sequence of the chimpanzee genome and comparison with the human genome** *Nature* **437**:69–87
11. Currat M., Excoffier L., Maddison W., Otto S.P., Ray N., Whitlock M.C., Yeaman S (2006) **Comment on “Ongoing adaptive evolution of ASPM, a brain size determinant in Homo sapiens” and “Microcephalin, a gene regulating brain size, continues to evolve adaptively in humans”** *Science* **313**
12. Danecek P., Auton A., Abecasis G., Albers C.A., Banks E., DePristo M.A., Handsaker R.E., Lunter G., Marth G.T., Sherry S.T., et al. (2011) **The variant call format and VCFtools** *Bioinformatics* **27**:2156–2158
13. Derrien T., Johnson R., Bussotti G., Tanzer A., Djebali S., Tilgner H., Guernec G., Martin D., Merkel A., Knowles D.G., et al. (2012) **The GENCODE v7 catalog of human long noncoding RNAs: analysis of their gene structure, evolution, and expression** *Genome Res* **22**:1775–1789
14. Dong X., Wang X., Zhang F., Tian W (2016) **Genome-Wide Identification of Regulatory Sequences Undergoing Accelerated Evolution in the Human Genome** *Mol Biol Evol* **33**:2565–2575

15. Evans P.D., Gilbert S.L., Mekel-Bobrov N., Vallender E.J., Anderson J.R., Vaez-Azizi L.M., Tishkoff S.A., Hudson R.R., Lahn B.T (2005) **Microcephalin, a gene regulating brain size, continues to evolve adaptively in humans** *Science* **309**:1717–1720
16. Fay J.C., Wu C.I (2000) **Hitchhiking under positive Darwinian selection** *Genetics* **155**:1405–1413
17. Fiddes I.T., Lodewijk G.A., Mooring M., Bosworth C.M., Ewing A.D., Mantalas G.L., Novak A.M., et al. (2018) **Human-Specific NOTCH2NL Genes Affect Notch Signaling and Cortical Neurogenesis** *Cell* **173**:1356–1369
18. Florio M., Heide M., Pinson A., Brandl H., Albert M., Winkler S., Wimberger P., Huttner W.B., Hiller M (2018) **Evolution and cell-type specificity of human-specific genes preferentially expressed in progenitors of fetal neocortex** *Elife* **7**
19. Franchini L.F., Pollard K.S (2017) **Human evolution: the non-coding revolution** *BMC Biol* **15**
20. GTEx Consortium (2017) **Genetic effects on gene expression across human tissues** *Nature* **550**:204–213
21. Hernando-Herraez I., Garcia-Perez R., Sharp A.J., Marques-Bonet T (2015) **DNA Methylation: Insights into Human Evolution** *PLoS Genet* **11**
22. Hodge R.D., Bakken T.E., Miller J.A., Smith K.A., Barkan E.R., Graybuck L.T., Close J.L., Long B., Johansen N., Penn O., et al. (2019) **Conserved cell types with divergent features in human versus mouse cortex** *Nature* **573**:61–68
23. Johnson A.D (2017) **The rewiring of transcription circuits in evolution** *Curr Opin Genet Dev* **47**:121–127
24. Johnson M.B., Sun X., Kodani A., Borges-Monroy R., Girsakis K.M., Ryu S.C., Wang P.P., Patel K., Gonzalez D.M., Woo Y.M., et al. (2018) **Aspm knockout ferret reveals an evolutionary mechanism governing cerebral cortical size** *Nature* **556**:370–375
25. Kaessmann H (2010) **Origins, evolution, and phenotypic impact of new genes** *Genome Res* **20**:1313–1326
26. Katoh K., Standley D.M (2013) **MAFFT Multiple Sequence Alignment Software Version 7: Improvements in Performance and Usability** *Mol Biol Evol* **30**:772–780
27. Krieger G., Lupo O., Wittkopp P., Barkai N (2022) **Evolution of transcription factor binding through sequence variations and turnover of binding sites** *Genome Res* **32**:1099–1111
28. Kuhn R.M., Haussler D., Kent W.J (2013) **The UCSC genome browser and associated tools** *Brief Bioinform* **14**:144–161
29. Kumar S., Stecher G., Tamura K. (2016) **MEGA7: Molecular Evolutionary Genetics Analysis Version 7.0 for Bigger Datasets** *Mol Biol Evol* **33**
30. Lee J.T (2009) **Lessons from X-chromosome inactivation: long ncRNA as guides and tethers to the epigenome** *Genes Dev* **23**:1831–1842

31. Li M., Santpere G., Imamura Kawasawa Y., Evgrafov O.V., Gulden F.O., Pochareddy S., Sunkin S.M., Li Z., Shin Y., Zhu Y., et al. (2018) **Integrative functional genomic analysis of human brain development and neuropsychiatric risks** *Science* **362**
32. Lin J., Wen Y., He S., Yang X., Zhang H., Zhu H (2019) **Pipelines for cross-species and genome-wide prediction of long noncoding RNA binding** *Nat Protoc* **14**:795–818
33. Liu J., Mosti F., Silver D.L (2021) **Human brain evolution: Emerging roles for regulatory DNA and RNA** *Curr Opin Neurobiol* **71**:170–177
34. Mangan R.J., Alsina F.C., Mosti F., Sotelo-Fonseca J.E., Snellings D.A., Au E.H., Carvalho J., Sathyan L., Johnson G.D., Reddy T.E., et al. (2022) **Adaptive sequence divergence forged new neurodevelopmental enhancers in humans** *Cell* **185**:4587–4603
35. McCoy R.C., Wakefield J., Akey J.M (2017) **Impacts of Neanderthal-Introgressed Sequences on the Landscape of Human Gene Expression** *Cell* **168**
36. McLean C.Y., Reno P.L., Pollen A.A., Bassan A.I., Capellini T.D., Guenther C., Indjeian V.B., Lim X., Menke D.B., Schaar B.T., et al. (2011) **Human-specific loss of regulatory DNA and the evolution of human-specific traits** *Nature* **471**:216–219
37. Mekel-Bobrov N., Gilbert S.L., Evans P.D., Vallender E.J., Anderson J.R., Hudson R.R., Tishkoff S.A., Lahn B.T (2005) **Ongoing adaptive evolution of ASPM, a brain size determinant in Homo sapiens** *Science* **309**:1720–1722
38. Meyer M., Kircher M., Gansauge M.T., Li H., Racimo F., Mallick S., Schraiber J.G., Jay F., Prufer K., de Filippo C., et al. (2012) **A High-Coverage Genome Sequence from an Archaic Denisovan Individual** *Science* **338**:222–226
39. Mondal T., Subhash S., Vaid R., Enroth S., Uday S., Reinius B., Mitra S., Mohammed A., James A.R., Hoberg E., et al. (2015) **MEG3 long noncoding RNA regulates the TGF-beta pathway genes through formation of RNA-DNA triplex structures** *Nat Commun* **6**
40. Nawrocki E.P (2014) **Annotating functional RNAs in genomes using Infernal** *Methods Mol Biol* **1097**:163–197
41. Nawrocki E.P., Eddy S.R (2013) **Infernal 1.1: 100-fold faster RNA homology searches** *Bioinformatics* **29**:2933–2935
42. Nawrocki E.P., Kolbe D.L., Eddy S.R (2009) **Infernal 1.0: inference of RNA alignments** *Bioinformatics* **25**:1335–1337
43. Nielsen R., Slatkin M. (2013) **Nielsen, R., and Slatkin, M. (2013). An Introduction to Population Genetics. Sinauer Associates Inc Publishers. An Introduction to Population Genetics. Sinauer Associates Inc Publishers**
44. Otto W., Stadler P.F., Lopez-Giraldez F., Townsend J.P., Lynch V.J., Wagner G.P (2009) **Measuring transcription factor-binding site turnover: a maximum likelihood approach using phylogenies** *Genome Biol Evol* **1**:85–98
45. Pinson A., Xing L., Namba T., Kalebic N., Peters J., Oegema C.E., Traikov S., Reppe K., Riesenberger S., Maricic T., et al. (2022) **Human TKTL1 implies greater neurogenesis in frontal neocortex of modern humans than Neanderthals** *Science* **377**

46. Pollen A.A., Bhaduri A., Andrews M.G., Nowakowski T.J., Meyerson O.S., Mostajo-Radji M.A., Di Lullo E., Alvarado B., Bedolli M., Dougherty M.L., et al. (2019) **Establishing Cerebral Organoids as Models of Human-Specific Brain Evolution** *Cell* **176**:743–756
47. Pollen A.A., Kilik U., Lowe C.B., Camp J.G (2023) **Human-specific genetics: new tools to explore the molecular and cellular basis of human evolution** *Nat Rev Genet*
48. Prabhakar S., Noonan J.P., Paabo S., Rubin E.M (2006) **Accelerated evolution of conserved noncoding sequences in humans** *Science* **314**
49. Prud'homme B., Gompel N., Carroll S.B (2007) **Emerging principles of regulatory evolution** *Proc Natl Acad Sci U S A* **104**:8605–8612
50. Prufer K., de Filippo C., Grote S., Mafessoni F., Korlevic P., Hajdinjak M., Vernot B., Skov L., Hsieh P.S., Peyregne S., et al. (2017) **A high-coverage Neandertal genome from Vindija Cave in Croatia** *Science* **358**:655–658
51. Prufer K., Munch K., Hellmann I., Akagi K., Miller J.R., Walenz B., Koren S., Sutton G., Kodira C., Winer R., et al. (2012) **The bonobo genome compared with the chimpanzee and human genomes** *Nature* **486**:527–531
52. Prufer K., Racimo F., Patterson N., Jay F., Sankararaman S., Sawyer S., Heinze A., Renaud G., Sudmant P.H., de Filippo C., et al. (2013) **The complete genome sequence of a Neanderthal from the Altai Mountains** *Nature* **505**:43–49
53. Purcell S., Neale B., Todd-Brown K., Thomas L., Ferreira M.A., Bender D., Maller J., Sklar P., de Bakker P.I., Daly M.J., et al. (2007) **PLINK: a tool set for whole-genome association and population-based linkage analyses** *Am J Hum Genet* **81**:559–575
54. Raudvere U., Kolberg L., Kuzmin I., Arak T., Adler P., Peterson H., Vilo J (2019) **g:Profiler: a web server for functional enrichment analysis and conversions of gene lists (2019 update)** *Nucleic Acids Res* **47**:W191–W198
55. Reilly S.K., Yin J., Ayoub A.E., Emera D., Leng J., Cotney J., Sarro R., Rakic P., Noonan J.P (2015) **Evolutionary genomics. Evolutionary changes in promoter and enhancer activity during human corticogenesis** *Science* **347**:1155–1159
56. Epigenomics Roadmap, Kundaje C., Meuleman A., Ernst W., Bilenky J., Yen M., Heravi-Moussavi A., Kheradpour A., Zhang P., Wang Z., et al. (2015) **Integrative analysis of 111 reference human epigenomes** *Nature* **518**:317–330
57. Sarropoulos I., Marin R., Cardoso-Moreira M., Kaessmann H (2019) **Developmental dynamics of lncRNAs across mammalian organs and species** *Nature* **571**:510–514
58. Slatkin M (2008) **Linkage disequilibrium--understanding the evolutionary past and mapping the medical future** *Nat Rev Genet* **9**:477–485
59. Suzuki I.K., Gacquer D., Van Heurck R., Kumar D., Wojno M., Bilheu A., Herpoel A., Lambert N., Cheron J., Polleux F., et al. (2018) **Human-Specific NOTCH2NL Genes Expand Cortical Neurogenesis through Delta/Notch Regulation** *Cell* **173**:1370–1384
60. Tajima F (1989) **Statistical method for testing the neutral mutation hypothesis by DNA polymorphism** *Genetics* **123**:585–595

61. Timpson N., Heron J., Smith G.D., Enard W (2007) **Comment on papers by Evans et al. and Mekel-Bobrov et al. on Evidence for Positive Selection of MCPH1 and ASPM.** *Science* **317**
62. Vilella A.J., Blanco-Garcia A., Hutter S., Rozas J (2005) **VariScan: Analysis of evolutionary patterns from large-scale DNA sequence polymorphism data** *Bioinformatics* **21**:2791–2793
63. Weir B.S., Cockerham C.C (1984) **Estimating F-Statistics for the Analysis of Population Structure** *Evolution* **38**:1358–1370
64. Wen Y., Wu Y., Xu B., Lin J., Zhu H (2022) **Fasim-LongTarget enables fast and accurate genome-wide lncRNA/DNA binding prediction** *Comput Struct Biotechnol J* **20**:3347–3350
65. West J.A., Davis C.P., Sunwoo H., Simon M.D., Sadreyev R.I., Wang P.I., Tolstorukov M.Y., Kingston R.E (2014) **The long noncoding RNAs NEAT1 and MALAT1 bind active chromatin sites** *Mol Cell* **55**:791–802
66. Whalen S., Pollard K.S (2022) **Enhancer Function and Evolutionary Roles of Human Accelerated Regions** *Annu Rev Genet* **56**:423–439
67. Xu C., Li Q., Efimova O., He L., Tatsumoto S., Stepanova V., Oishi T., Udono T., Yamaguchi K., Shigenobu S., et al. (2018) **Human-specific features of spatial gene expression and regulation in eight brain regions** *Genome Res* **28**:1097–1110
68. Yu F., Hill R.S., Schaffner S.F., Sabeti P.C., Wang E.T., Mignault A.A., Ferland R.J., Moyzis R.K., Walsh C.A., Reich D (2007) **Comment on “Ongoing adaptive evolution of ASPM, a brain size determinant in Homo sapiens”** *Science* **316**
69. Yue F., Cheng Y., Breschi A., Vierstra J., Wu W., Ryba T., Sandstrom R., Ma Z., Davis C., Pope B.D., et al. (2014) **A comparative encyclopedia of DNA elements in the mouse genome** *Nature* **515**:355–364
70. Zhang X., Fang B., Huang Y.F (2023) **Transcription factor binding sites are frequently under accelerated evolution in primates** *Nat Commun* **14**
71. Zhu Y., Sousa A.M.M., Gao T., Skarica M., Li M., Santpere G., Esteller-Cucala P., Juan D., Ferrandez-Peral L., Gulden F.O., et al. (2018) **Spatiotemporal transcriptomic divergence across human and macaque brain development** *Science* **362**
1. Agoglia R.M., Sun D., Birey F., Yoon S.J., Miura Y., Sabatini K., Pasca S.P., Fraser H.B (2021) **Primate cell fusion disentangles gene regulatory divergence in neurodevelopment** *Nature* **592**:421–427
2. Akbari A., Vitti J.J., Iranmehr A., Bakhtiari M., Sabeti P.C., Mirarab S., Bafna V (2018) **Identifying the favored mutation in a positive selective sweep** *Nat Methods* **15**:279–282
3. Chen H., Patterson N., Reich D (2010) **Population differentiation as a test for selective sweeps** *Genome Res* **20**:393–402
4. Ernst J., Kheradpour P., Mikkelsen T.S., Shores N., Ward L.D., Epstein C.B., Zhang X., Wang L., Issner R., Coyne M., et al. (2011) **Mapping and analysis of chromatin state dynamics in nine human cell types** *Nature* **473**:43–49

5. Fan S., Kelly D.E., Beltrame M.H., Hansen M.E.B., Mallick S., Ranciaro A., Hirbo J., Thompson S., Beggs W., Nyambo T., et al. (2019) **African evolutionary history inferred from whole genome sequence data of 44 indigenous African populations** *Genome Biol* **20**
6. Fay J.C., Wu C.I (2000) **Hitchhiking under positive Darwinian selection** *Genetics* **155**:1405–1413
7. Florio M., Heide M., Pinson A., Brandl H., Albert M., Winkler S., Wimberger P., Huttner W.B., Hiller M (2018) **Evolution and cell-type specificity of human-specific genes preferentially expressed in progenitors of fetal neocortex** *Elife* **7**
8. Giri A.K., Bharadwaj S., Banerjee P., Chakraborty S., Parekatt V., Rajashekar D., Tomar A., Ravindran A., Basu A., Tandon N., et al. (2017) **DNA methylation profiling reveals the presence of population-specific signatures correlating with phenotypic characteristics** *Mol Genet Genomics* **292**:655–662
9. GTEx Consortium (2017) **Genetic effects on gene expression across human tissues** *Nature* **550**:204–213
10. He S., Zhang H., Liu H., Zhu H (2015) **LongTarget: a tool to predict lncRNA DNA-binding motifs and binding sites via Hoogsteen base-pairing analysis** *Bioinformatics* **31**:178–186
11. Johnson M.B., Sun X., Kodani A., Borges-Monroy R., Girsakis K.M., Ryu S.C., Wang P.P., Patel K., Gonzalez D.M., Woo Y.M., et al. (2018) **Aspm knockout ferret reveals an evolutionary mechanism governing cerebral cortical size** *Nature* **556**:370–375
12. Li M., Santpere G., Imamura Kawasawa Y., Evgrafov O.V., Gulden F.O., Pochareddy S., Sunkin S.M., Li Z., Shin Y., Zhu Y., et al. (2018) **Integrative functional genomic analysis of human brain development and neuropsychiatric risks** *Science* **362**
13. Mathieson I., Lazaridis I., Rohland N., Mallick S., Patterson N., Roodenberg S.A., Harney E., Stewardson K., Fernandes D., Novak M., et al. (2015) **Genome-wide patterns of selection in 230 ancient Eurasians** *Nature* **528**:499–503
14. Meissner A., Mikkelsen T.S., Gu H., Wernig M., Hanna J., Sivachenko A., Zhang X., Bernstein B.E., Nusbaum C., Jaffe D.B., et al. (2008) **Genome-scale DNA methylation maps of pluripotent and differentiated cells** *Nature* **454**:766–770
15. Meyer M., Kircher M., Gansauge M.T., Li H., Racimo F., Mallick S., Schraiber J.G., Jay F., Prufer K., de Filippo C., et al. (2012) **A High-Coverage Genome Sequence from an Archaic Denisovan Individual** *Science* **338**:222–226
16. Mondal T., Subhash S., Vaid R., Enroth S., Uday S., Reinius B., Mitra S., Mohammed A., James A.R., Hoberg E., et al. (2015) **MEG3 long noncoding RNA regulates the TGF-beta pathway genes through formation of RNA-DNA triplex structures** *Nat Commun* **6**
17. Ng B., White C.C., Klein H.U., Sieberts S.K., McCabe C., Patrick E., Xu J., Yu L., Gaiteri C., Bennett D.A., et al. (2017) **An xQTL map integrates the genetic architecture of the human brain's transcriptome and epigenome** *Nat Neurosci* **20**:1418–1426
18. Pollen A.A., Bhaduri A., Andrews M.G., Nowakowski T.J., Meyerson O.S., Mostajo-Radji M.A., Di Lullo E., Alvarado B., Bedolli M., Dougherty M.L., et al. (2019) **Establishing Cerebral Organoids as Models of Human-Specific Brain Evolution** *Cell* **176**:743–756

19. Reilly S.K., Yin J., Ayoub A.E., Emera D., Leng J., Cotney J., Sarro R., Rakic P., Noonan J.P (2015) **Evolutionary genomics. Evolutionary changes in promoter and enhancer activity during human corticogenesis** *Science* **347**:1155–1159
20. Slatkin M (2008) **Linkage disequilibrium--understanding the evolutionary past and mapping the medical future** *Nat Rev Genet* **9**:477–485
21. Suzuki I.K., Gacquer D., Van Heurck R., Kumar D., Wojno M., Bilheu A., Herpoel A., Lambert N., Cheron J., Polleux F., et al. (2018) **Human-Specific NOTCH2NL Genes Expand Cortical Neurogenesis through Delta/Notch Regulation** *Cell* **173**:1370–1384
22. Tajima F (1989) **Statistical method for testing the neutral mutation hypothesis by DNA polymorphism** *Genetics* **123**:585–595
23. Weir B.S., Cockerham C.C (1984) **Estimating F-Statistics for the Analysis of Population Structure** *Evolution* **38**:1358–1370
24. Wen Y., Wu Y., Xu B., Lin J., Zhu H (2022) **Fasim-LongTarget enables fast and accurate genome-wide lncRNA/DNA binding prediction** *Comput Struct Biotechnol J* **20**:3347–3350
25. West J.A., Davis C.P., Sunwoo H., Simon M.D., Sadreyev R.I., Wang P.I., Tolstorukov M.Y., Kingston R.E (2014) **The long noncoding RNAs NEAT1 and MALAT1 bind active chromatin sites** *Mol Cell* **55**:791–802
26. Xu C., Li Q., Efimova O., He L., Tatsumoto S., Stepanova V., Oishi T., Udono T., Yamaguchi K., Shigenobu S., et al. (2018) **Human-specific features of spatial gene expression and regulation in eight brain regions** *Genome Res* **28**:1097–1110
27. Ye C.J., Feng T., Kwon H.K., Raj T., Wilson M.T., Asinowski N., McCabe C., Lee M.H., Frohlich I., Paik H.I., et al. (2014) **Intersection of population variation and autoimmunity genetics in human T cell activation** *Science* **345**
28. Zhu Y., Sousa A.M.M., Gao T., Skarica M., Li M., Santpere G., Esteller-Cucala P., Juan D., Ferrandez-Peral L., Gulden F.O., et al. (2018) **Spatiotemporal transcriptomic divergence across human and macaque brain development** *Science* **362**

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Reviewer #1 (Public Review):

Summary

While DNA sequence divergence, differential expression, and differential methylation analysis have been conducted between humans and the great apes to study changes that "make us human", the role of lncRNAs and their impact on the human genome and biology has not been fully explored. In this study, the authors computationally predict HSlncRNAs as well as their DNA Binding sites using a method they have developed previously and then examine these predicted regions with different types of enrichment analyses. Broadly, the analysis is straightforward and after identifying these regions/HSlncRNAs the authors examined their effects using different external datasets.

Strengths/weaknesses

By and large, the analysis performed is dependent on their ability to identify HSlncRNAs and their DBS. I think that they have done a good job of showing the performance metrics of their methods in previous publications. Thereafter, they perform a series of enrichment-type analyses that have been used in the field for quite a while now to look at tissue-specific enrichment, or region-specific enrichment, or functional enrichment, and I think these have been carried out well. The authors achieved the aims of their work. I think one of the biggest contributions that this paper brings to the field is their annotation of these HSlncRNAs. Thus a major revisionary effort could be spent on applying their method to the latest genomes that have been released so that the community could get a clean annotation of newly identified HSlncRNAs (see comment 2).

Comments

1. Though some of their results about certain HSlncRNAs having DBSs in all genes is rather surprising/suspicious, I think that broadly their process to identify and validate DBSs is robust, they have multiple lines of checks to identify such regions, including functional validation. These predictions are bound to have some level of false positive/negative rate and it might be nice to restate those here and on what experiment/validation data these were conducted. However, the rest of their analysis comprises different types of enrichment analysis which shouldn't be affected by outlier HSlncRNAs if indeed their FPR/FNR are low.
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4. Isn't the correct control to examine whether eQTLs are more enriched in HSlncRNA DBSs a set of transcription factor binding sites? I don't think using just promoter regions is a reasonable control here. This does not take away from the broader point however that eQTLs are found in DBSs and I think they can perform this alternate test.
5. In the discussion, they highlight the evolution of sugar intake, which I'm not sure is appropriate. This comes not from GO enrichment but rather from a few genes that are found at the tail of their distribution. While these signals may be real, the evolution of traits is often highly polygenic and they don't see this signal in their functional enrichment. I suggest removing that line. Moreover, HSlncRNAs are ones that are unique across a much longer time frame than the transition to agriculture which is when sugar intake rose greatly. Thus, it's unlikely to see enrichment for something that arose in the past 6000-7000 years would in the annotation that is designed to detect human-chimp or human-neanderthal level divergence.

Reviewer #2 (Public Review):

Lin et al attempt to examine the role of lncRNAs in human evolution in this manuscript. They apply a suite of population genetics and functional genomics analyses that leverage existing data sets and public tools, some of which were previously built by the authors, who clearly have experience with lncRNA binding prediction. However, I worry that there is a lack of suitable methods and/or relevant controls at many points and that the interpretation is too quick to infer selection. While I don't doubt that lnc RNAs contribute to the evolution of

modern humans, and certainly agree that this is a question worth asking, I think this paper would benefit from a more rigorous approach to tackling it.

At this point, my suggestions are mostly focused on tightening and strengthening the methods; it is hard for me to predict the consequence of these changes on the results or their interpretation, but as a general rule I also encourage the authors to not over-interpret their conclusions in terms of what phenotype was selected for when as they do at certain points (eg glucose metabolism).

I note some specific points that I think would benefit from more rigorous approaches, and suggest possible ways forward for these.

1. Much of this work is focused on comparing DNA binding domains in human-unique long-noncoding RNAs and DNA binding sites across the promoters of genes in the human genome, and I think the authors can afford to be a bit more methodical/selective in their processing and filtering steps here. The article begins by searching for orthologues of human lncRNAs to arrive at a set of 66 human-specific lncRNAs, which are then characterised further through the rest of the manuscript. Line 99 describes a binding affinity metric used to separate strong DBS from weak DBS; the methods (line 432) describe this as being the product of the DBS or lncRNA length times the average Identity of the underlying TTSs. This multiplication, in fact, undoes the standardising value of averaging and introduces a clear relationship between the length of a region being tested and its overall score, which in turn is likely to bias all downstream inference, since a long lncRNA with poor average affinity can end up with a higher score than a short one with higher average affinity, and it's not quite clear to me what the biological interpretation of that should be. Why was this metric defined in this way?

2. There is also a strong assumption that identified sites will always be bound (line 100), which I disagree is well-supported by additional evidence (lines 109-125). The authors show that predicted NEAT1 and MALAT1 DBS overlap experimentally validated sites for NEAT1, MALAT1, and MEG3, but this is not done systematically, or genome-wide, so it's hard to know if the examples shown are representative, or a best-case scenario.

It's also not quite clear how overlapping promoters or TSS are treated - are these collapsed into a single instance when calculating genome-wide significance? If, eg, a gene has five isoforms, and these differ in the 3' UTR but their promoter region contains a DBS, is this counted five times, or one? Since the interaction between the lncRNA and the DBS happens at the DNA level, it seems like not correcting for this uneven distribution of transcripts is likely to skew results, especially when testing against genome-wide distributions, eg in the results presented in sections 5 and 6. I do not think that comparing genes and transcripts putatively bound by the 40 HS lncRNAs to a random draw of 10,000 lncRNA/gene pairs drawn from the remaining ~13500 lncRNAs that are not HS is a fair comparison. Rather, it would be better to do many draws of 40 non-HS lncRNAs and determine an empirical null distribution that way, if possible actively controlling for the overall number of transcripts (also see the following point).

3. Thresholds for statistical testing are not consistent, or always well justified. For instance, in line 142 GO testing is performed on the top 2000 genes (according to different rankings), but there's no description of the background regions used as controls anywhere, or of why 2000 genes were chosen as a good number to test? Why not 1000, or 500? Are the results overall robust to these (and other) thresholds? Then line 190 the threshold for downstream testing is now the top 20% of genes, etc. I am not opposed to different thresholds in principle, but they should be justified.

Likewise, comparing Tajima's D values near promoters to genome-wide values is unfair, because promoters are known to be under strong evolutionary constraints relative to background regions; as such it is not surprising that the results of this comparison are

significant. A fairer comparison would attempt to better match controls (eg to promoters without HS lncRNA DBS, which I realise may be nearly impossible), or generate empirical p-values via permutation or simulation.

4. There are huge differences in the comparisons between the Vindija and Altai Neanderthal genomes that to me suggest some sort of technical bias or the such is at play here. e.g. line 190 reports 1256 genes to have a high distance between the Altai Neanderthal and modern humans, but only 134 Vindija genes reach the same cutoff of 0.034. The temporal separation between the two specimens does not seem sufficient to explain this difference, nor the difference between the Altai Denisovan and Neanderthal results (2514 genes for Denisovan), which makes me wonder if it is a technical artefact relating to the quality of the genome builds? It would be worth checking.

5. Inferring evolution: There are some points of the manuscript where the authors are quick to infer positive selection. I would caution that GTEx contains a lot of different brain tissues, thus finding a brain eQTL is a lot easier than finding a liver eQTL, just because there are more opportunities for it. Likewise, claims in the text and in Tables 1 and 2 about the evolutionary pressures underlying specific genes should be more carefully stated. The same is true when the authors observe high F_{st} between groups (line 515), which is only one possible cause of high F_{st} - population differentiation and drift are just as capable of giving rise to it, especially at small sample sizes.

Author Response

Reviewer #1 (Public Review):

Summary

While DNA sequence divergence, differential expression, and differential methylation analysis have been conducted between humans and the great apes to study changes that "make us human", the role of lncRNAs and their impact on the human genome and biology has not been fully explored. In this study, the authors computationally predict HS lncRNAs as well as their DNA Binding sites using a method they have developed previously and then examine these predicted regions with different types of enrichment analyses. Broadly, the analysis is straightforward and after identifying these regions/HS lncRNAs the authors examined their effects using different external datasets.

Strengths/weaknesses

By and large, the analysis performed is dependent on their ability to identify HS lncRNAs and their DBS. I think that they have done a good job of showing the performance metrics of their methods in previous publications. Thereafter, they perform a series of enrichment-type analyses that have been used in the field for quite a while now to look at tissue-specific enrichment, or region-specific enrichment, or functional enrichment, and I think these have been carried out well. The authors achieved the aims of their work. I think one of the biggest contributions that this paper brings to the field is their annotation of these HS lncRNAs. Thus a major revisionary effort could be spent on applying their method to the latest genomes that have been released so that the community could get a clean annotation of newly identified HS lncRNAs (see comment 2).

Comments

1. Though some of their results about certain HSIncRNAs having DBSs in all genes is rather surprising/suspicious, I think that broadly their process to identify and validate DBSs is robust, they have multiple lines of checks to identify such regions, including functional validation. These predictions are bound to have some level of false positive/negative rate and it might be nice to restate those here and on what experiment/validation data these were conducted. However, the rest of their analysis comprises different types of enrichment analysis which shouldn't be affected by outlier HSIncRNAs if indeed their FPR/FNR are low.

1. There are now several new genomes available as part of the Zoonomia consortium and 240 Primate consortium papers released. These papers have re-examined some annotations such as Human Accelerated Regions (HARs) and found with a larger dataset as well as better reference genomes, that a large fraction of HARs were actually incorrectly annotated - that is that they were also seen in other lineages outside of just the great apes. If these papers have not already examined HSIncRNAs, the authors should try and re-run the computational predictions with this updated set and then identify HSIncRNAs there. This might help to clarify their signal and remove lncRNAs that might be present in other primates but are somehow missing in the great apes. This might also help to mitigate some results that they see in section 3 of their paper in comparing DBS distances between archaics and humans.

1. The differences between the archaic hominins in their DBS distances to modern humans are a bit concerning. At some level, we expect these to be roughly similar when examining African modern humans and perhaps the Denisovan being larger when examining Europeans and Asians, but they seem to have distances that aren't expected given the demography. In addition, from their text for section 3, they begin by stating that they are computing two types of distances but then I lost track of which distance they were discussing in paragraph 3 of section 3. Explicitly stating which of the two distances in the text would be helpful for the reader.

(1) According to Figure 1A (according actually to Meyer et al., 2012, Prüfer et al., 2017, and Prüfer et al., 2013), the phylogenetic distance from modern humans to Denisovan is shorter than the distance to Altai Neanderthal. However, also according to these studies, the branch of Denisovan is more remote to modern humans than Altai Neanderthal. Thus, it is not unreasonable to find that 2514 and 1256 DBSs have distances > 0.034 in genes in Denisovans and Altai Neanderthals, respectively. Probably, both the phylogenetic distances and DBS distances depend considerably on the sampled genomes of Altai and Denisovan who lived on the earth for quite long. When new samples are obtained, these distances may be somewhat changed.

(2) Regarding “they are computing two types of distances but then I lost track of which distance they were discussing in paragraph 3 of section 3”, the second type of distances were discussed in section 3, and the distances computed in the first way were not further analyzed because “This defect may be caused by that the human ancestor was built using six primates without archaic humans”.

1. Isn't the correct control to examine whether eQTLs are more enriched in HSIncRNA DBSs a set of transcription factor binding sites? I don't think using just promoter regions is a reasonable control here. This does not take away from the broader point however that eQTLs are found in DBSs and I think they can perform this alternate test.

Indeed, the TFs-TFBSs and lncRNAs-DBSs relationships are comparable, and which one contains more QTLs is an interesting question. In this sense, it is reasonable to use TFBSs as the control. However, for three reasons, we did not perform the comparison and use TFBSs as the control. First, most TFBSs are predicted by varied methods, making us concern the reliability of comparing two sets of predictions. Second, most QTLs in DBSs are mQTLs but most QTLs in TFBSs are eQTLs. Third, probably a greater portion of TFBSs than DBSs are not in promoters, and the time consumption of LongTarget made us unable to predict DBSs truly genome-wide. Nevertheless, this is an interesting question deserving further exploring.

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Multiple sugar metabolism-related pathways, including “glucose homeostasis” and “glucose metabolic process”, are found to be enriched only in Altai Neanderthal but not in chimpanzees (Figure 2). Indeed, HS lncRNAs are across a much longer time frame than the transition to agriculture. However, given that apes and monkeys know picking the ripe, sugar-rich fruits at the right time and place, we conjecture that archaic humans as hunter-gatherer could effectively explore natural sugars.

Reviewer #2 (Public Review):

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Length is an important metric of DBS, but it has a defect – a triplex of 100 bp may have 50% or 70% of nucleotides bound; in the two situations, the binding affinity of DBD and DBS is very different.

1. There is also a strong assumption that identified sites will always be bound (line 100), which I disagree is well-supported by additional evidence (lines 109-125). The authors show that predicted NEAT1 and MALAT1 DBS overlap experimentally validated sites for NEAT1, MALAT1, and MEG3, but this is not done systematically, or genome-wide, so it's hard to know if the examples shown are representative, or a best-case scenario.

More details are described in the citation Wen et al. 2022. We will put the sites into Supplementary Tables in the revised version.

It's also not quite clear how overlapping promoters or TSS are treated - are these collapsed into a single instance when calculating genome-wide significance? If, eg, a gene has five isoforms, and these differ in the 3' UTR but their promoter region contains a DBS, is this counted five times, or one? Since the interaction between the lncRNA and the DBS happens at the DNA level, it seems like not correcting for this uneven distribution of transcripts is likely to skew results, especially when testing against genome-wide distributions, eg in the results presented in sections 5 and 6. I do not think that comparing genes and transcripts putatively bound by the 40 HS lncRNAs to a random draw of 10,000 lncRNA/gene pairs drawn from the remaining ~13500 lncRNAs that are not HS is a fair comparison. Rather, it would be better to do many draws of 40 non-HS lncRNAs and determine an empirical null distribution that way, if possible actively controlling for the overall number of transcripts (also see the following point).

(1) If, say, three transcripts of a gene share the same promoter region (i.e., they have the same TSS) but differ only in 3'UTR, the promoter region was used to predict DBSs just for once. Otherwise, if the three transcripts have different TSS, the three promoter regions were used to predict DBSs.

(2) A gene may have many DBSs if it has many transcripts, or few ones if it has just a few transcripts. We did not correct for this uneven distribution of transcripts, because our GTEx analysis was on the transcript level; it is well recognized that transcripts of the same gene can be expressed in different tissues.

(3) We randomly sampled a pair of non-HS lncRNA and a transcript for 10000 times (i.e., 10000 pairs). It is a point that multiple draws of 40 non-HS lncRNAs should be made to make

the statistics more robust.

1. Thresholds for statistical testing are not consistent, or always well justified. For instance, in line 142 GO testing is performed on the top 2000 genes (according to different rankings), but there's no description of the background regions used as controls anywhere, or of why 2000 genes were chosen as a good number to test? Why not 1000, or 500? Are the results overall robust to these (and other) thresholds? Then line 190 the threshold for downstream testing is now the top 20% of genes, etc. I am not opposed to different thresholds in principle, but they should be justified.

The over-representation analysis using g:Profiler was performed taking the whole genome as the background. Analyzing more DBSs (especially weak DBSs) would generate more results, but the results could be less reliable. Thus, there is a trade-off between analyzing fewer DBSs with relatively high reliability and analyzing more DBSs with relatively low reliability. Inevitably, the handling of this trade-off is somewhat subjective, and to carefully compare the two classes of DBSs per can be an independent question. Although weak DBSs were not systematically analyzed, the results from the strong DBSs undoubtedly suggest that HS lncRNAs have contributed greatly to human evolution.

Likewise, comparing Tajima's D values near promoters to genome-wide values is unfair, because promoters are known to be under strong evolutionary constraints relative to background regions; as such it is not surprising that the results of this comparison are significant. A fairer comparison would attempt to better match controls (eg to promoters without HS lncRNA DBS, which I realise may be nearly impossible), or generate empirical p-values via permutation or simulation.

We examined Tajima's D in DBSs (Supplementary Figure 9) and in HS lncRNA genes (Supplementary Figure 18). In both cases, we compared the Tajima's D values with the genome-wide background.

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We used the same workflow (and the same cutoff 0.034) to analyze Vindija and Altai Neanderthal and Denisovan. If a smaller cutoff was used, one would see more Vindija genes. The question again is that there is a trade-off. Analyzing epigenome and epigenetic regulation in archaic genomes is an interesting direction, and much more studies are needed before more reasonably setting related parameters and cutoffs.

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